

Evaluating Resource Adequacy Using Hourly 40 Weather-Year Data Under High VRE Penetration for the Luzon, Visayas, and Mindanao Grids in Alignment with the Power Development Plan(2023-2050)

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Adviser:

Adonis Emmanuel DC. Tio

Outline

- 1. Problem Statement**
- 2. Power Demand and Supply Outlook**
- 3. Resource Adequacy Discussion**
- 4. Visualization**
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- 11. Remaining Tasks**
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The Problem

- **PDP targets:** 35% renewable by 2030, 50% by 2040
- Current planning does not account for **multi-year extreme weather patterns**
- Wind and solar can drop significantly for **extended periods** during challenging years
- **Risk:** Systematic underestimation of capacity needed for supply adequacy
- International studies show single weather-year analysis leads to **misrepresentation of investment needs**

Luzon Power Demand and Supply Outlook

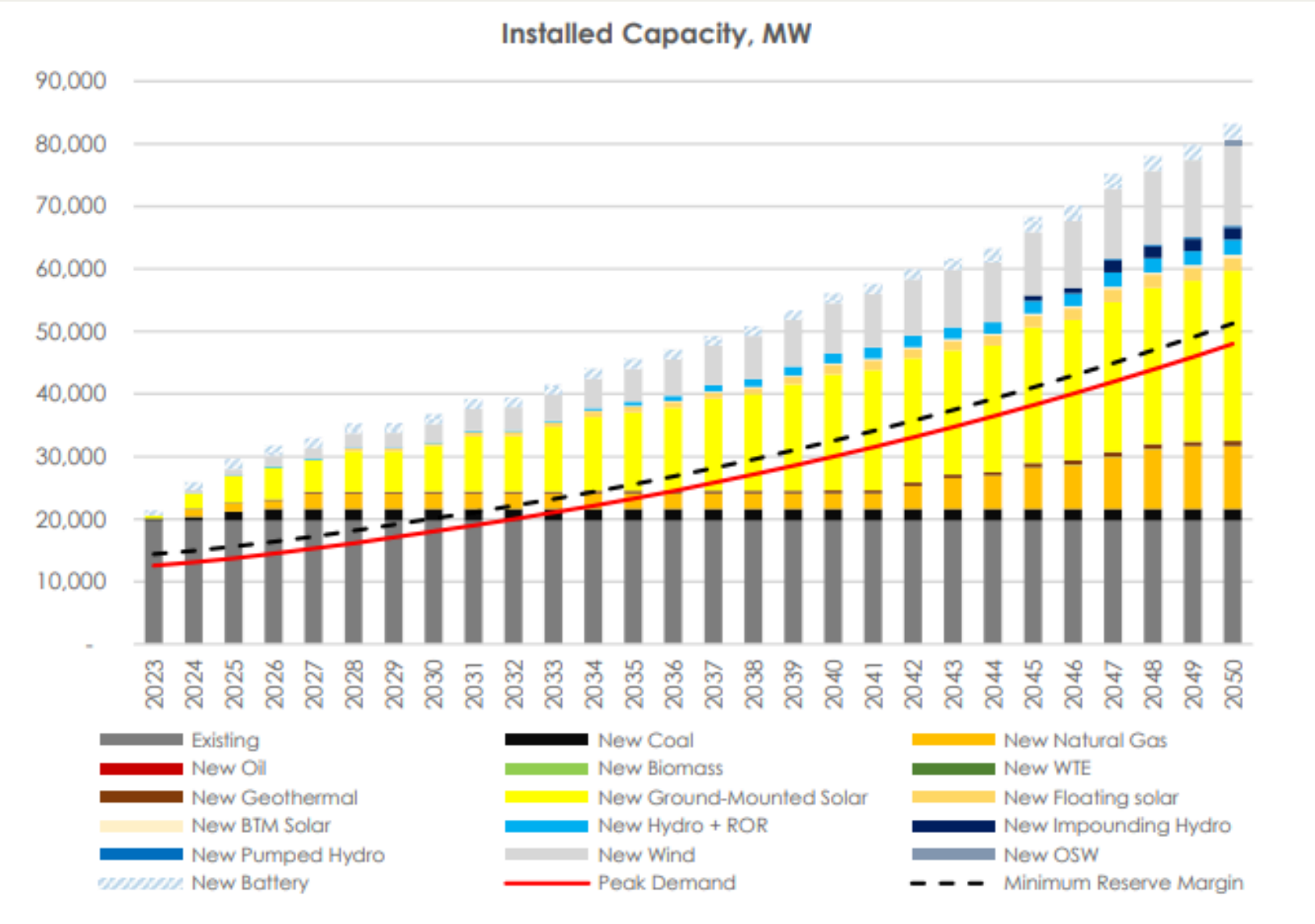


Figure 1.1: Reference Scenario

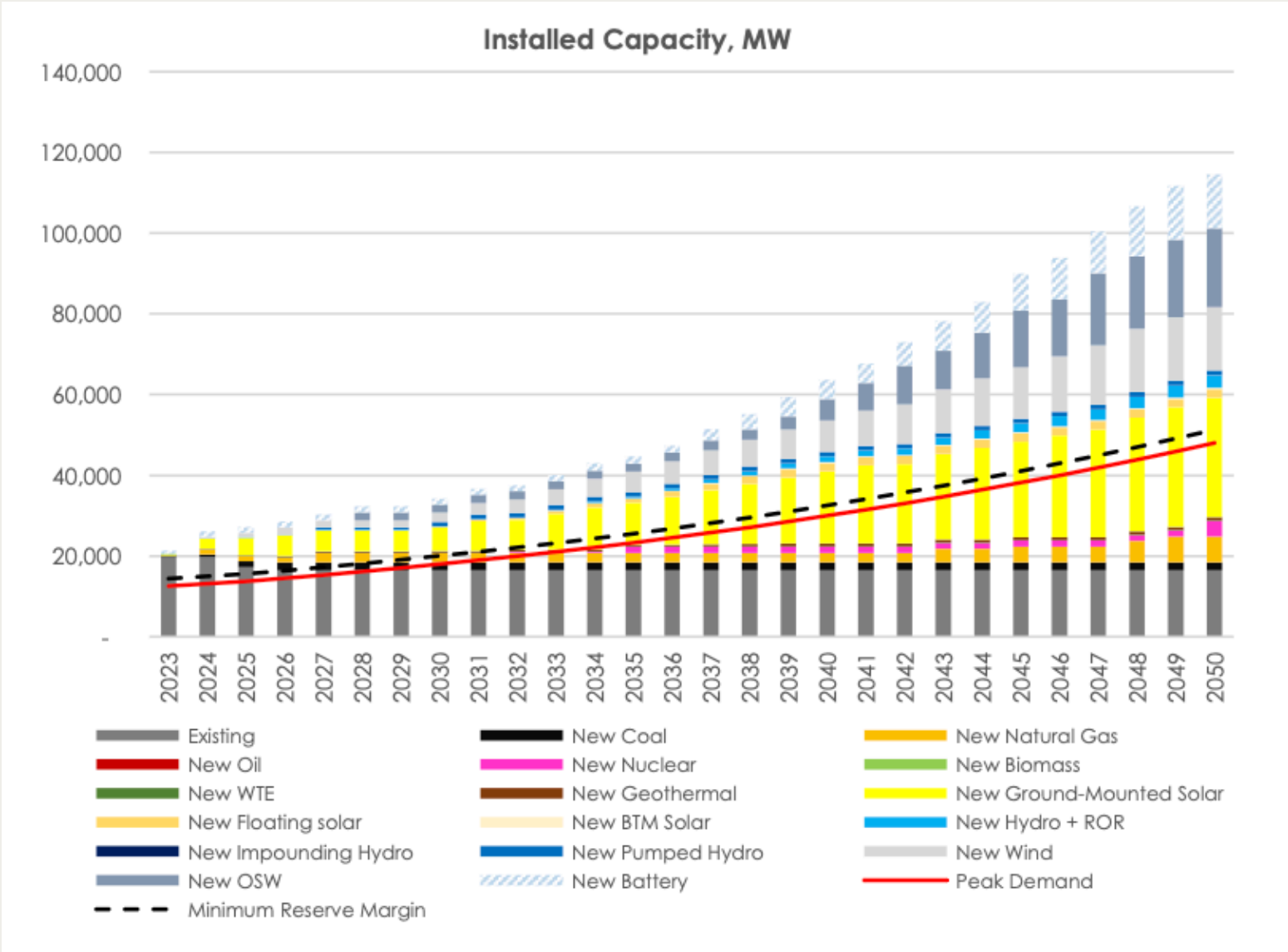


Figure 1.2: CES 1

Visayas Power Demand and Supply Outlook

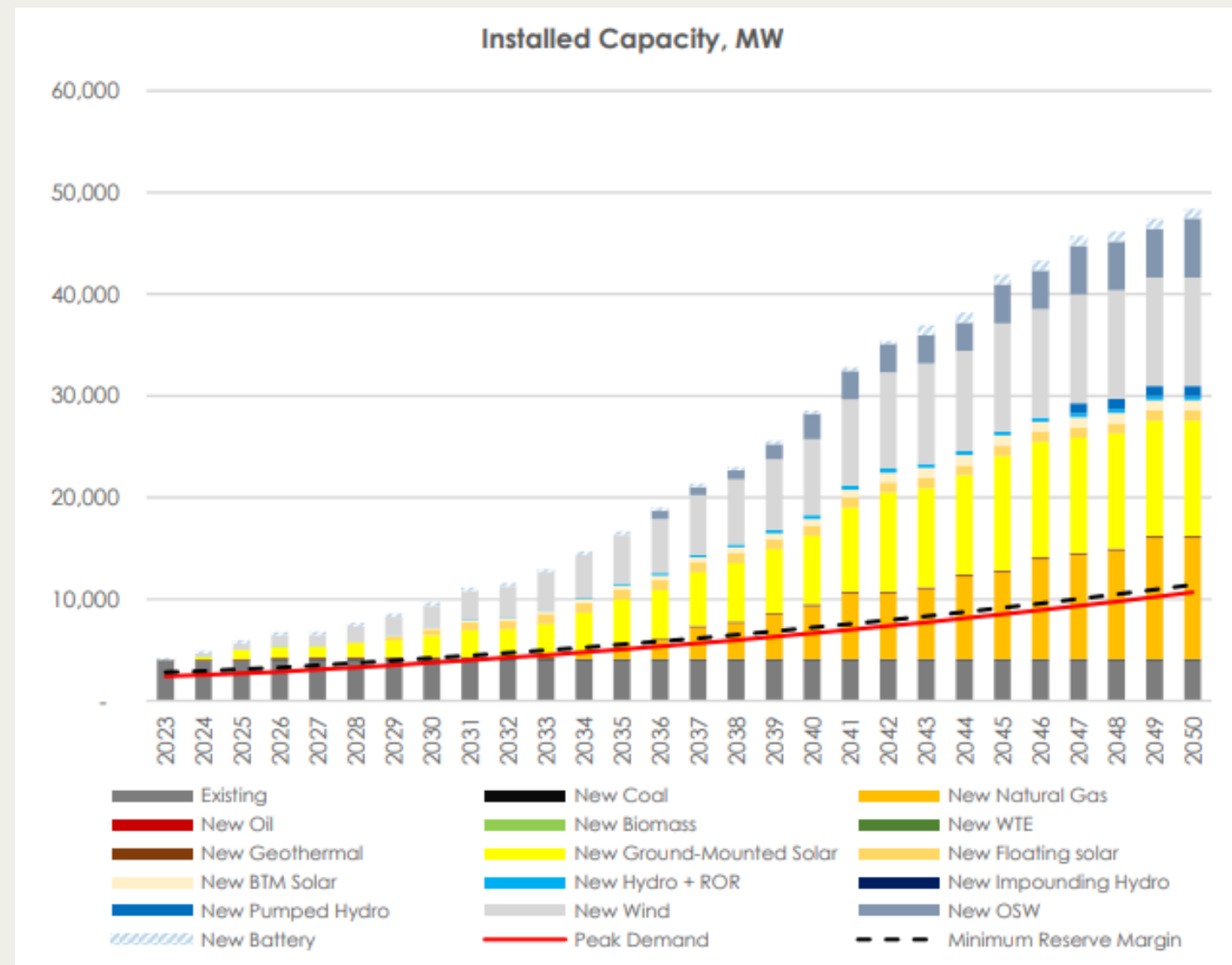


Figure 1.1: Reference Scenario

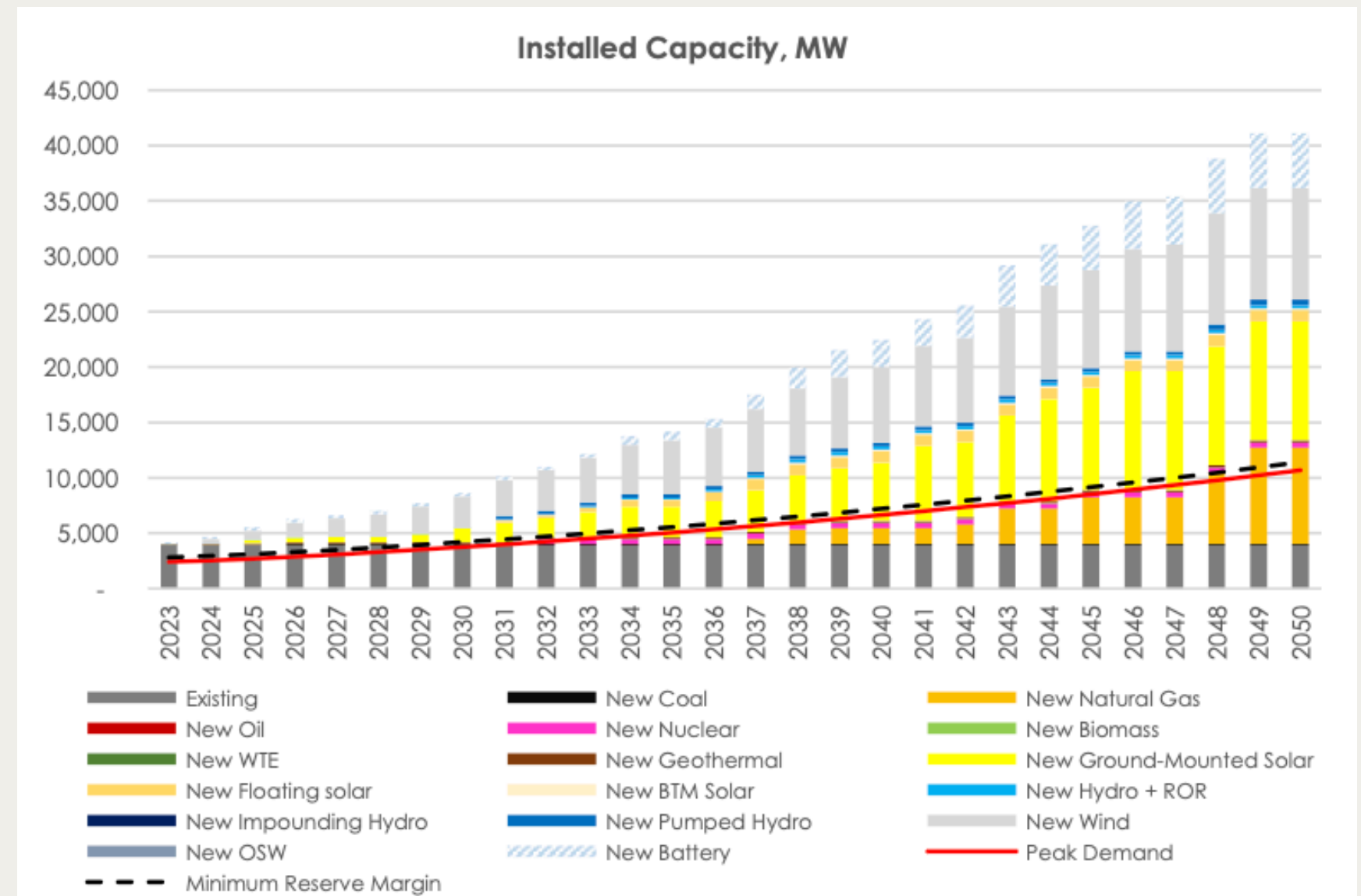


Figure 1.2: CES 1

Mindanao Power Demand and Supply Outlook

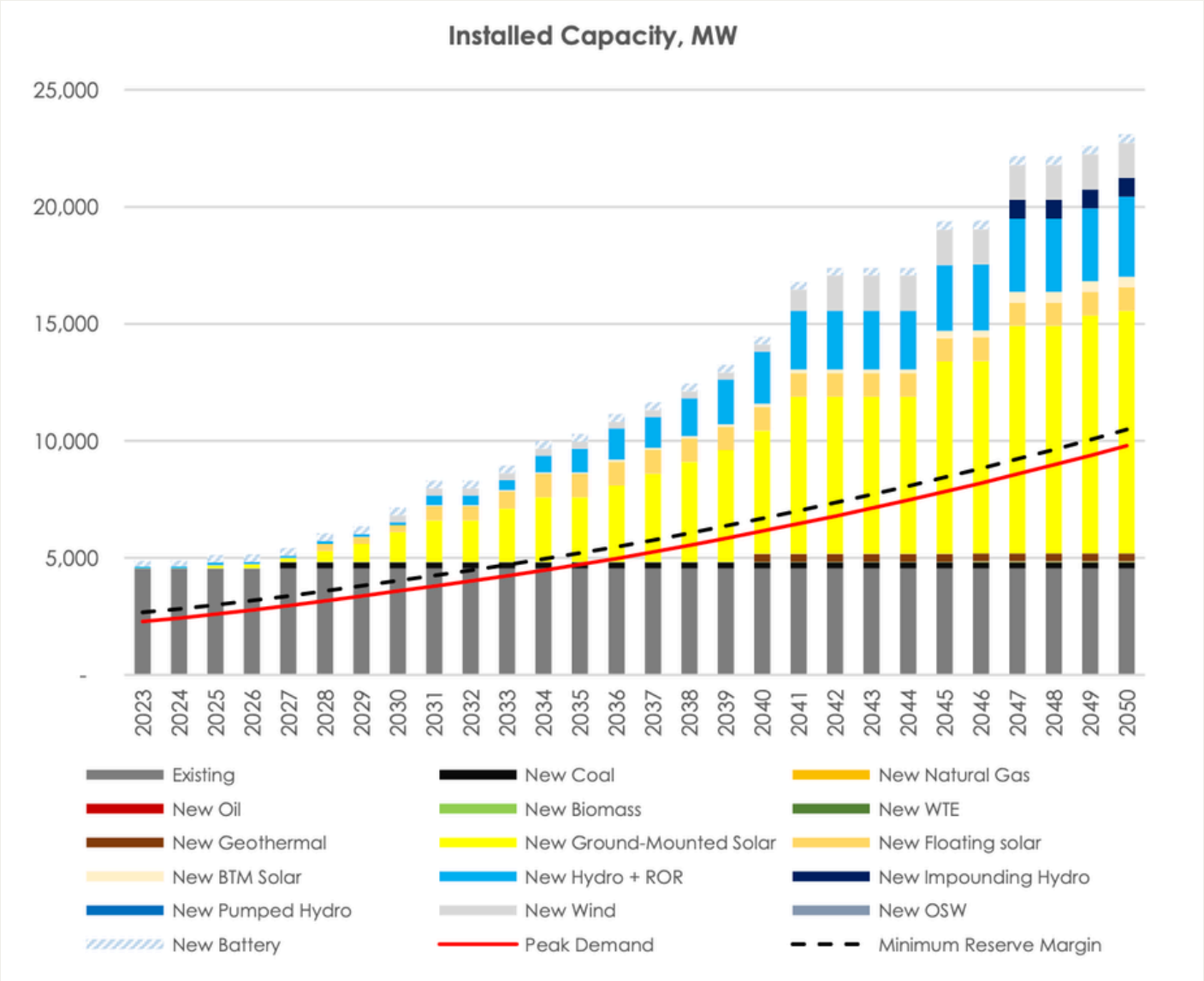


Figure 1.1: Reference Scenario

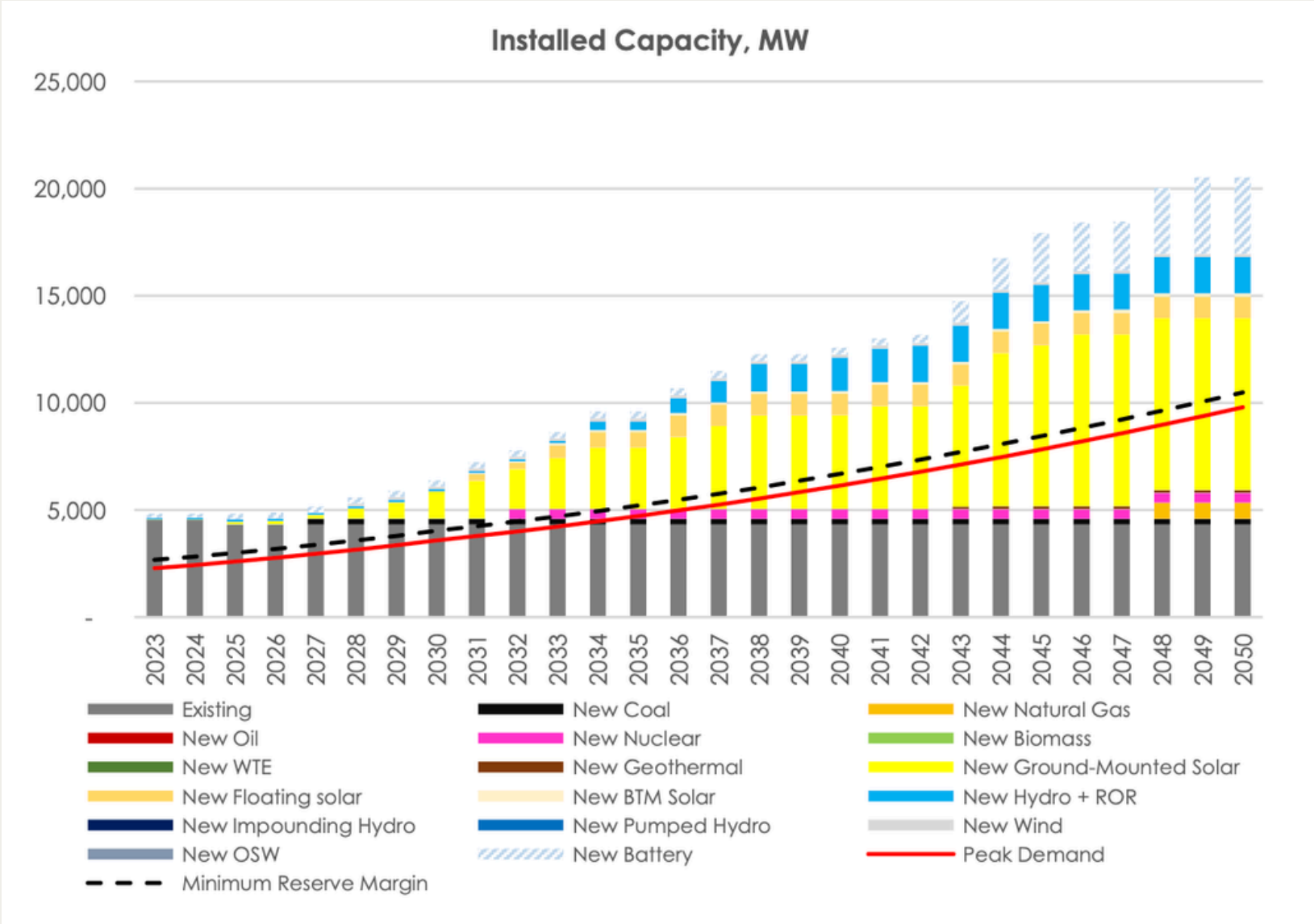


Figure 1.2: CES 1

Resource Adequacy

- **Definition:** Ability of supply-side and demand-side resources to meet aggregate electrical demand at all times
- **Metrics:** Loss of Load Expectation (LOLE), Expected Unserved Energy (EUE), etc.
- **Critical for:** Ensuring reliable service while balancing economic capacity investments

Visualization

Capacity Risk Zone

- The right of the vertical axis (red) will be called Capacity Risk Zone where the PDP lacks in dependable capacities.

Energy Risk Zone

- The values above the curve (orange) is the Energy Risk Zone where the batteries cannot handle such long-term events.

Safe Zone

- The area between the curve and the vertical axis (green) are the events that the dependable capacity targets can handle easily.

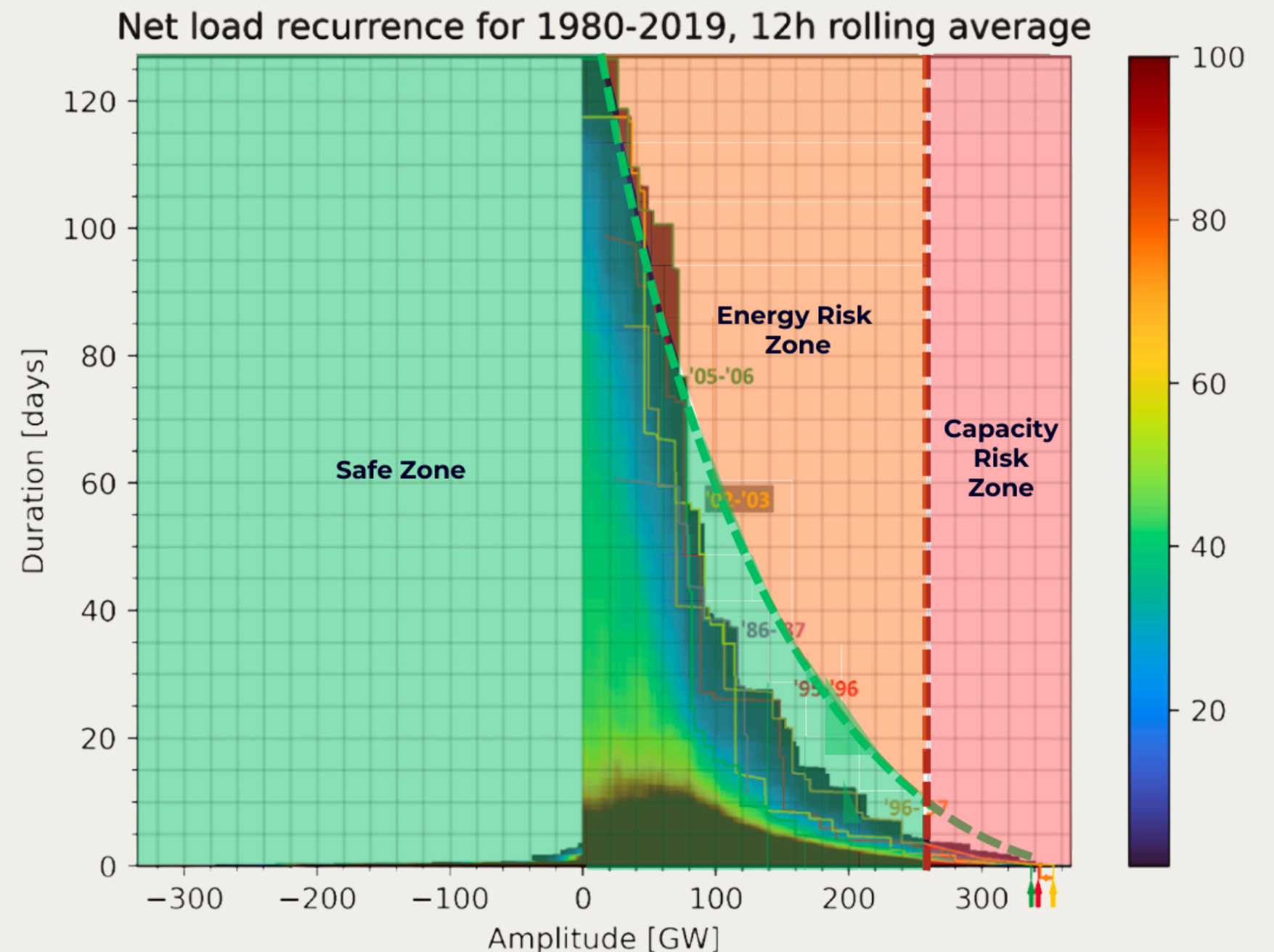


Figure 7: Figure 3 from Ullmark et al [7] with "Capacity Risk Zone", "Energy Risk Zone", and "Safe Zone" Determined

Project Objectives

The general objective of this study is to assess the sufficiency of the generation capacities of the Philippine power system under the capacity targets of the PDP 2023–2050 by characterizing the interannual variability of renewable energy resources. Specifically, this study aims to:

- a. Generate multi-decadal solar/wind profiles (40 years: 1985–2024)
- b. Characterize net load variability across all CREZ zones
- c. Evaluate adequacy of PDP projected capacities
- d. Visualize reliability risk zones on recurrence heatmaps

Scope and Limitations

Scope

- Luzon, Visayas, and Mindanao grids
- 2025–2050 (5-year intervals)
- 2 PDP scenarios (Reference Scenario and CES 1)

Limitations

- Single-node assumption (no transmission losses/congestion)
- MERRA-2 historical data only (no future climate change modeling)
- Spatial aggregation at CREZ centroids
- Battery storage: 4-hour fixed duration assumption

Methodology

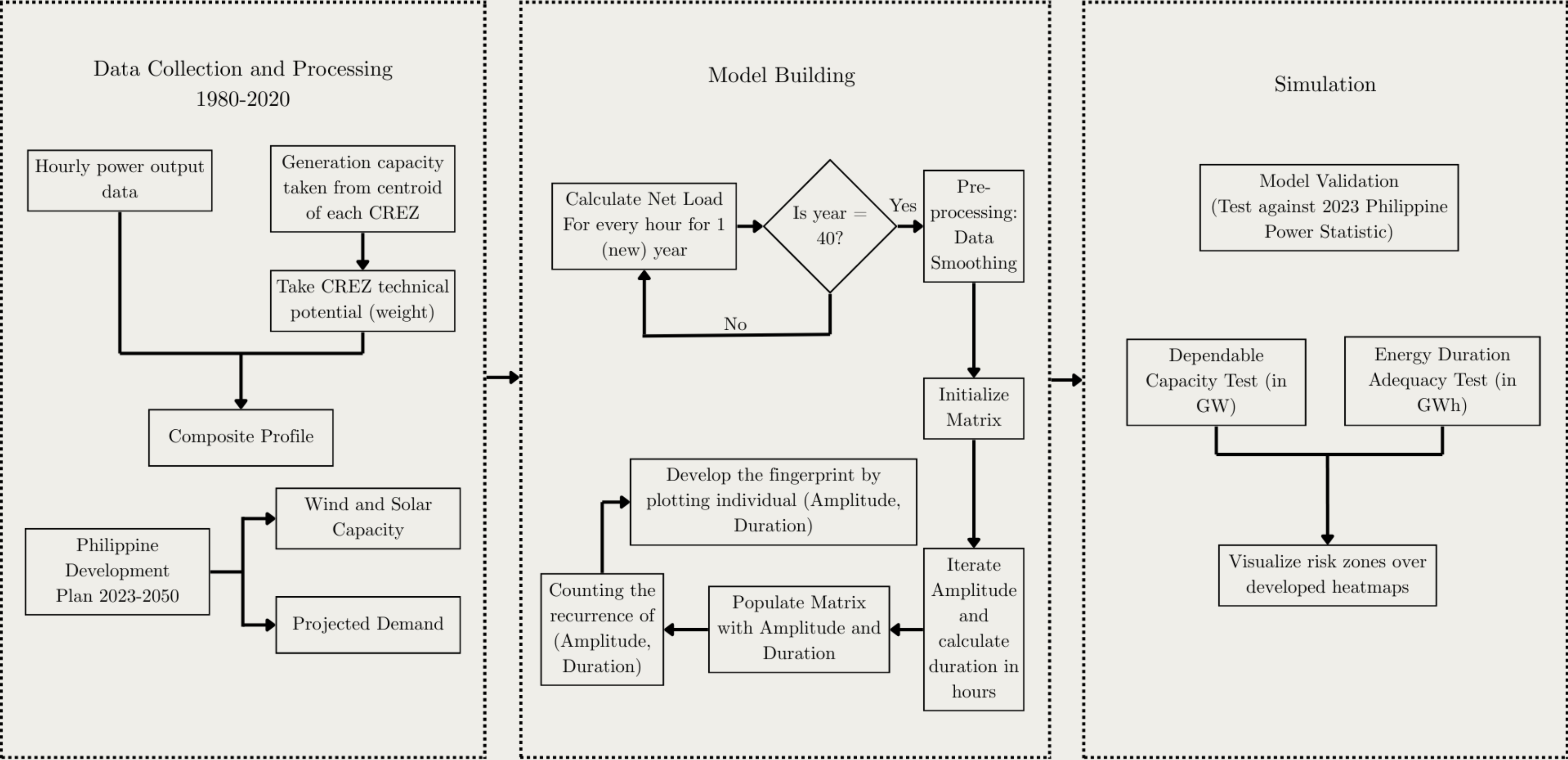


Figure 3: Methodology Overview Flowchart

Project Schedule and Deliverables

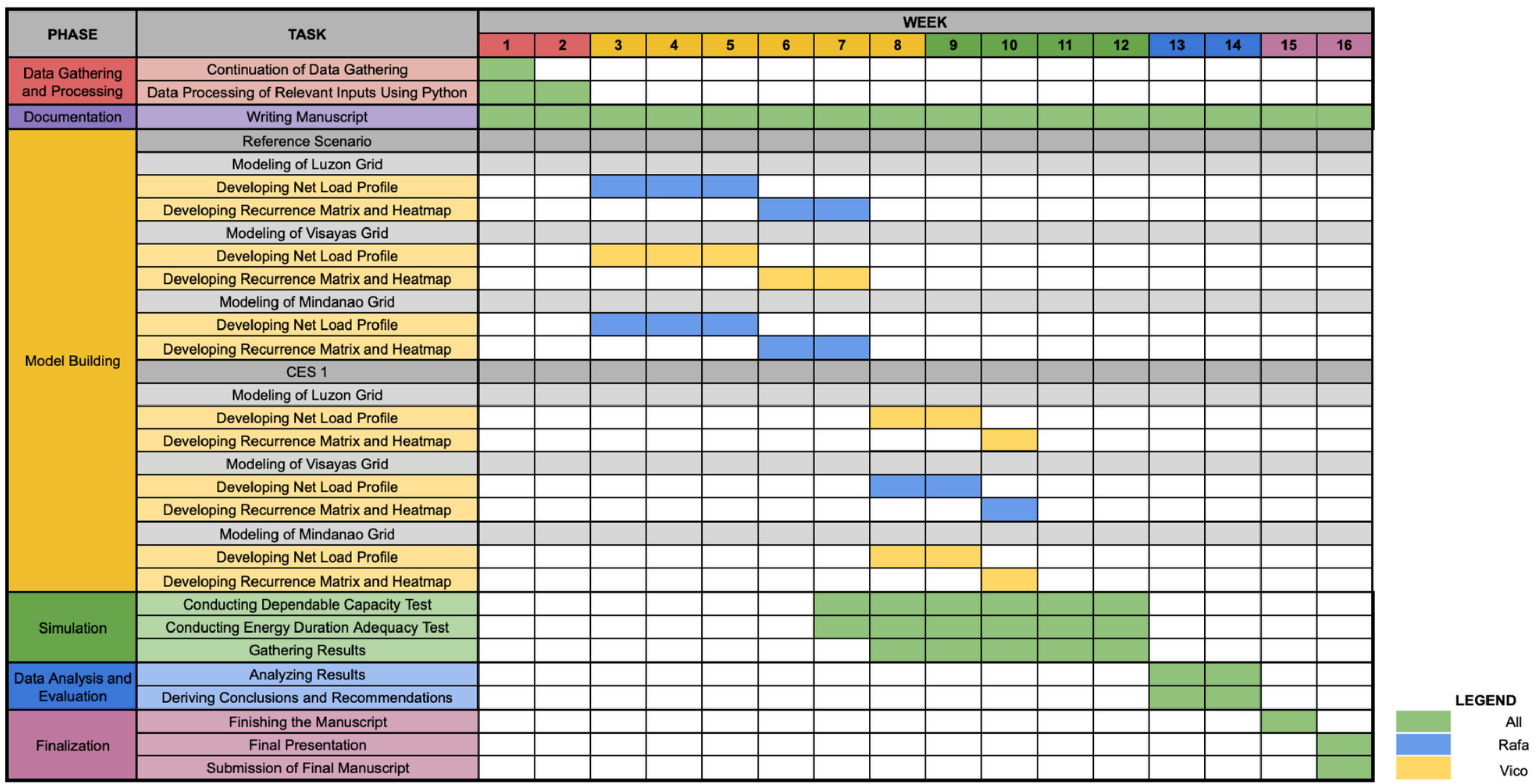


Figure 1: Gantt Chart

Project Schedule and Deliverables

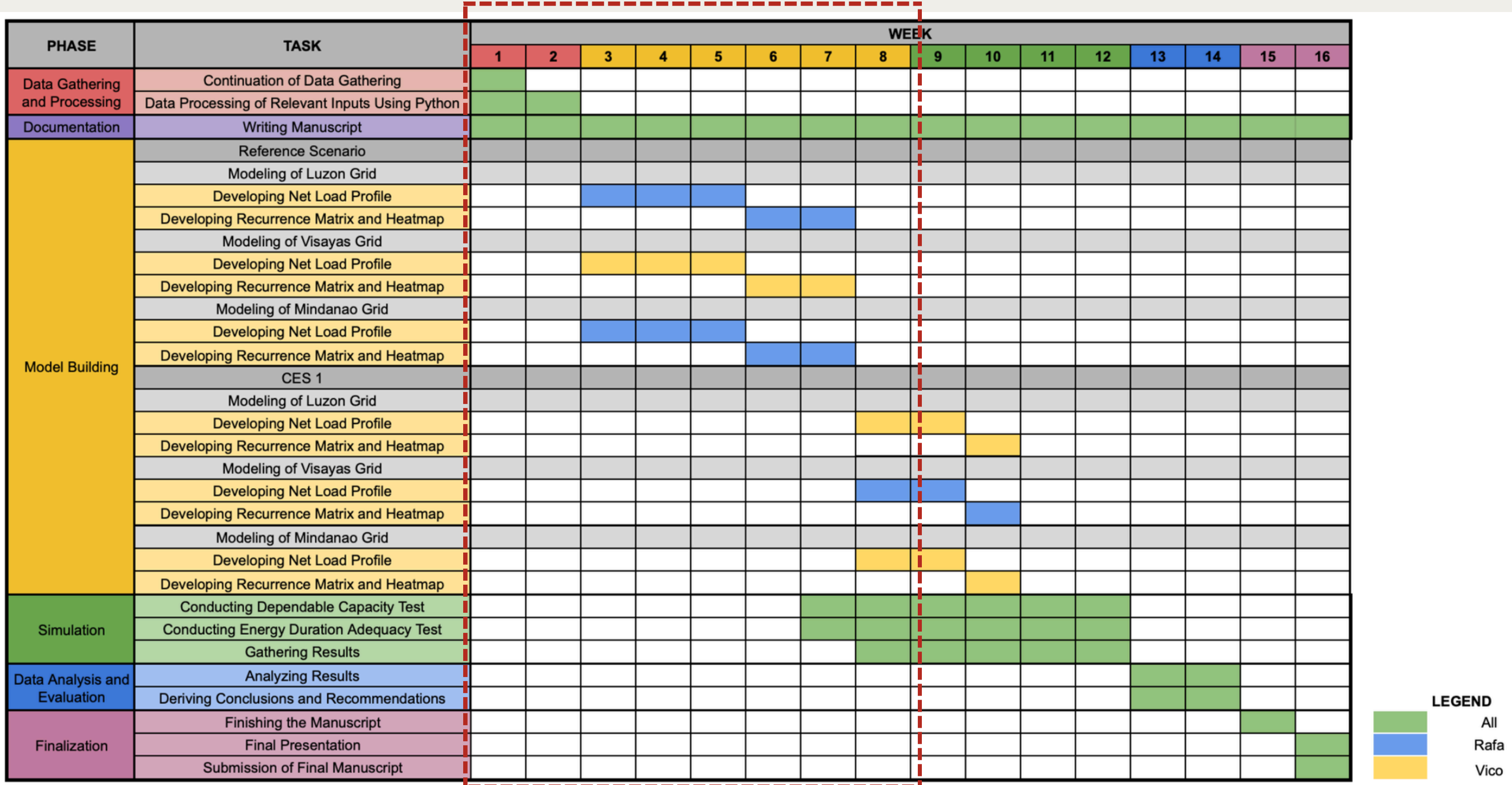


Figure 2: Gantt Chart – Halfway–point Deliverables

Project Schedule and Deliverables

Halfway-point Deliverables *(as of March 19, 2025)*

Halfway-point Deliverables		Status
1	Gather the data needed for Luzon, Visayas, and Mindanao grids	
2	Process the relevant inputs using Python.	
3	Develop net load profile of the grids individually for Reference Scenario.	
4	Develop recurrence matrix and heatmap of the grids individually for Reference Scenario.	
5	Start developing the net load profile of the grids individually for CES 1.	
6	Start conducting tests and gathering of results	

Pending

In progress

Completed

Data Repository

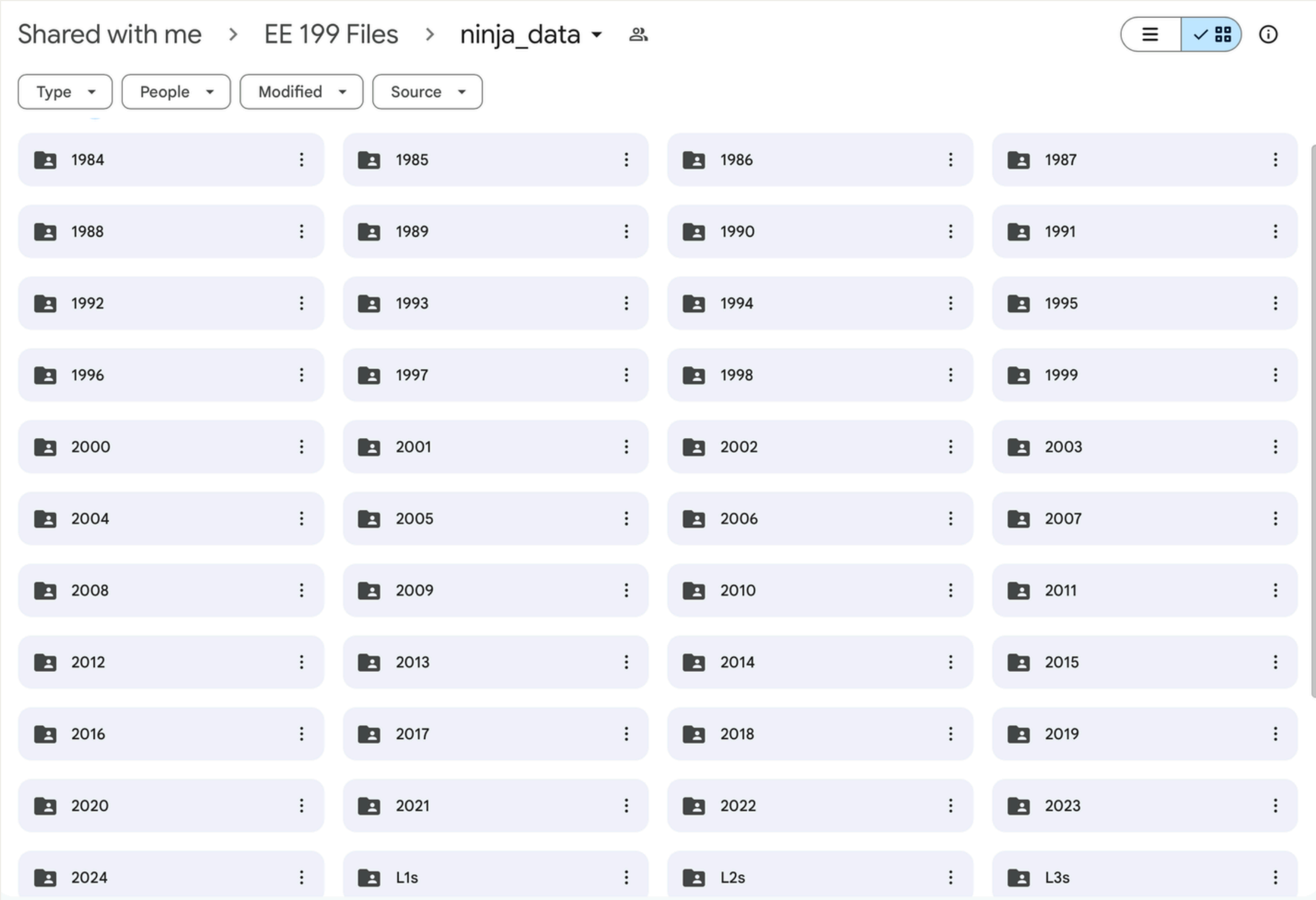
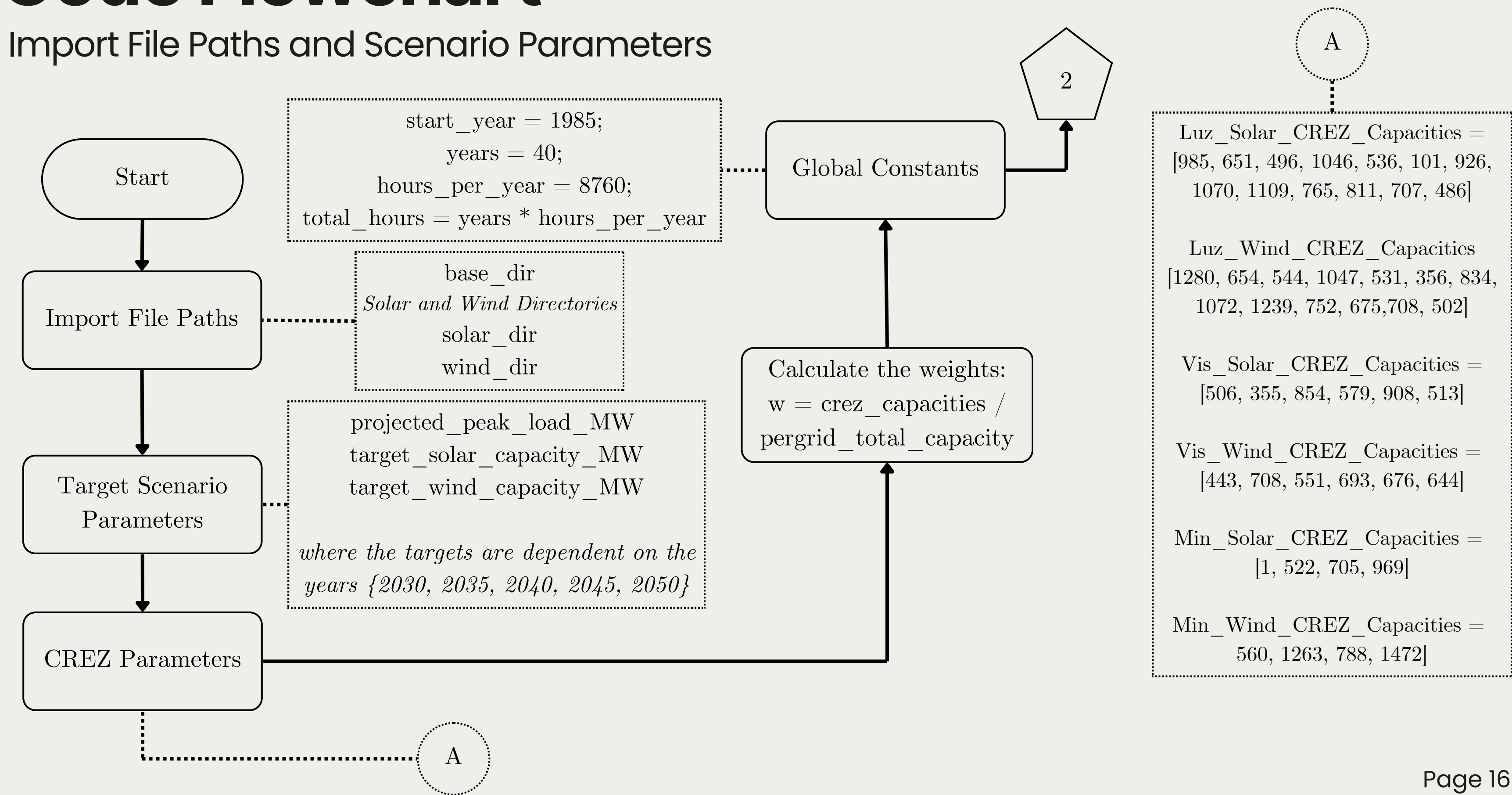


Figure 3: Data Repository

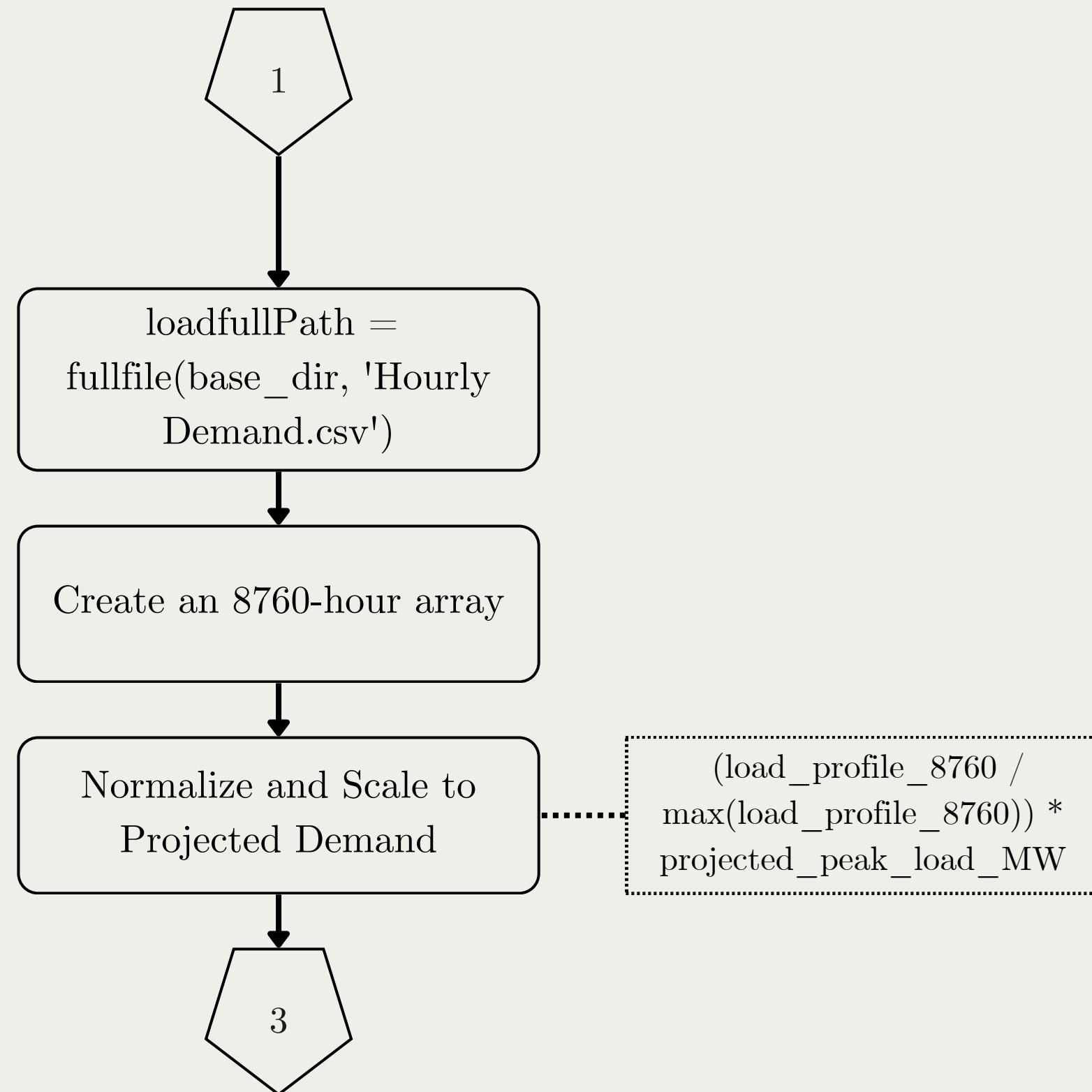
Code Flowchart

Import File Paths and Scenario Parameters



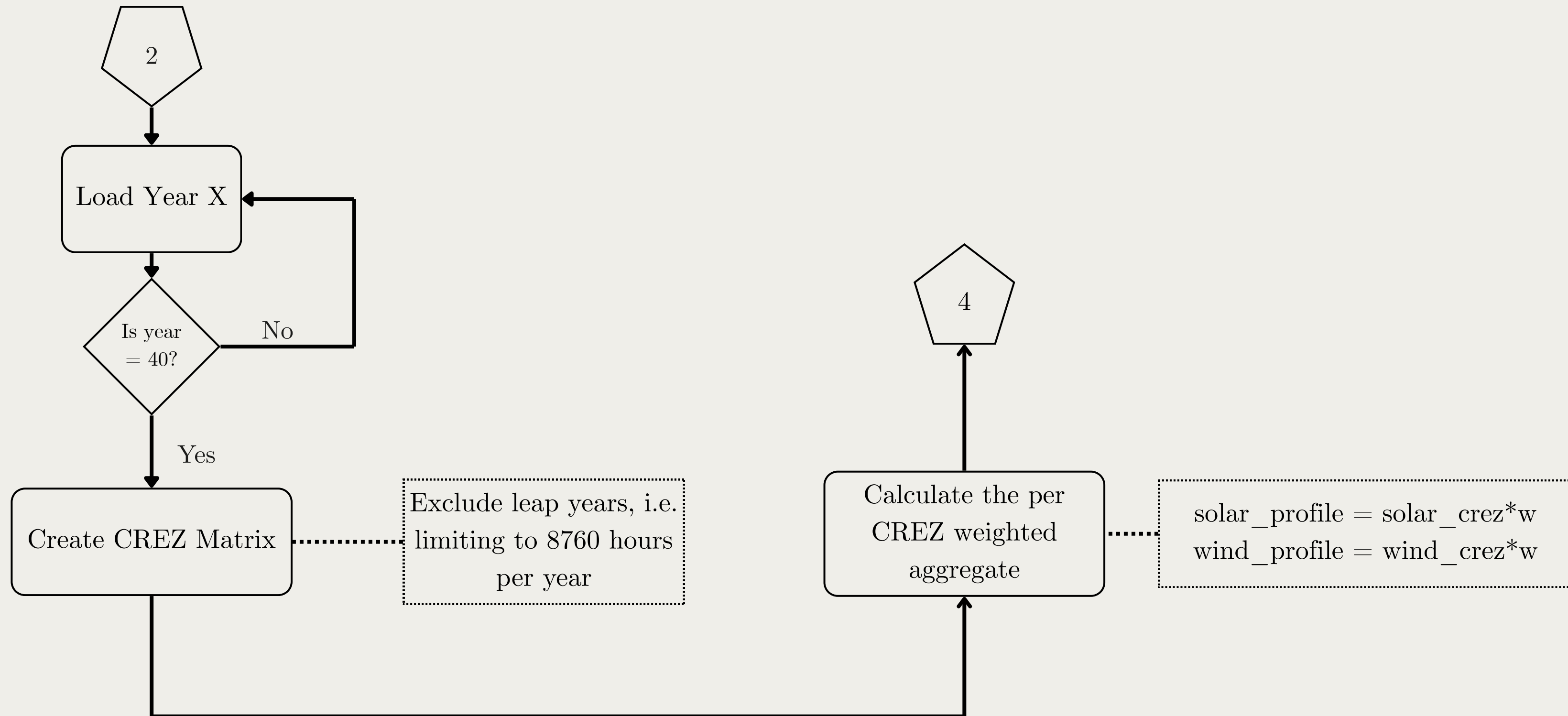
Code Flowchart

Import Load Behavior Profile



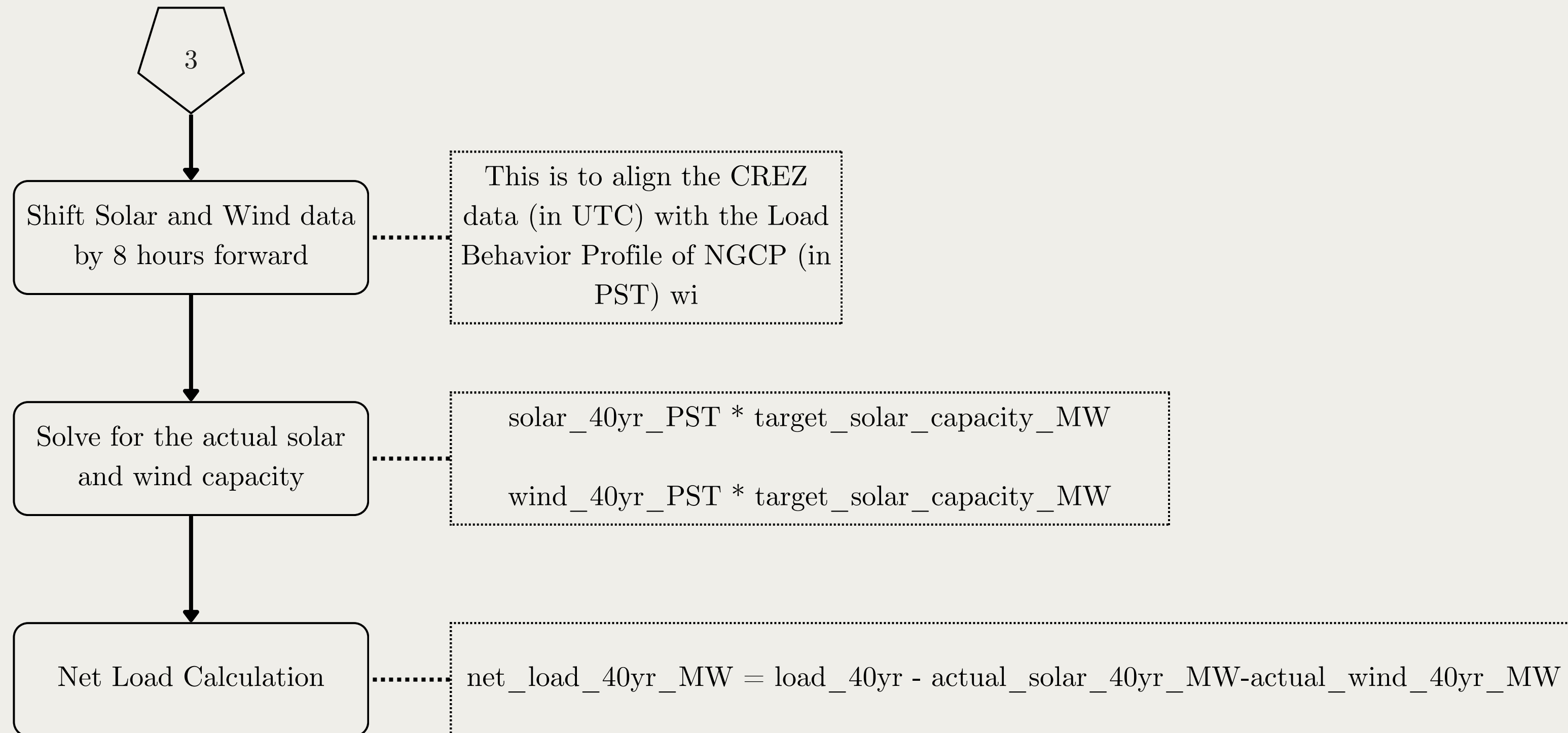
Code Flowchart

Import 40 Years of CREZ Data



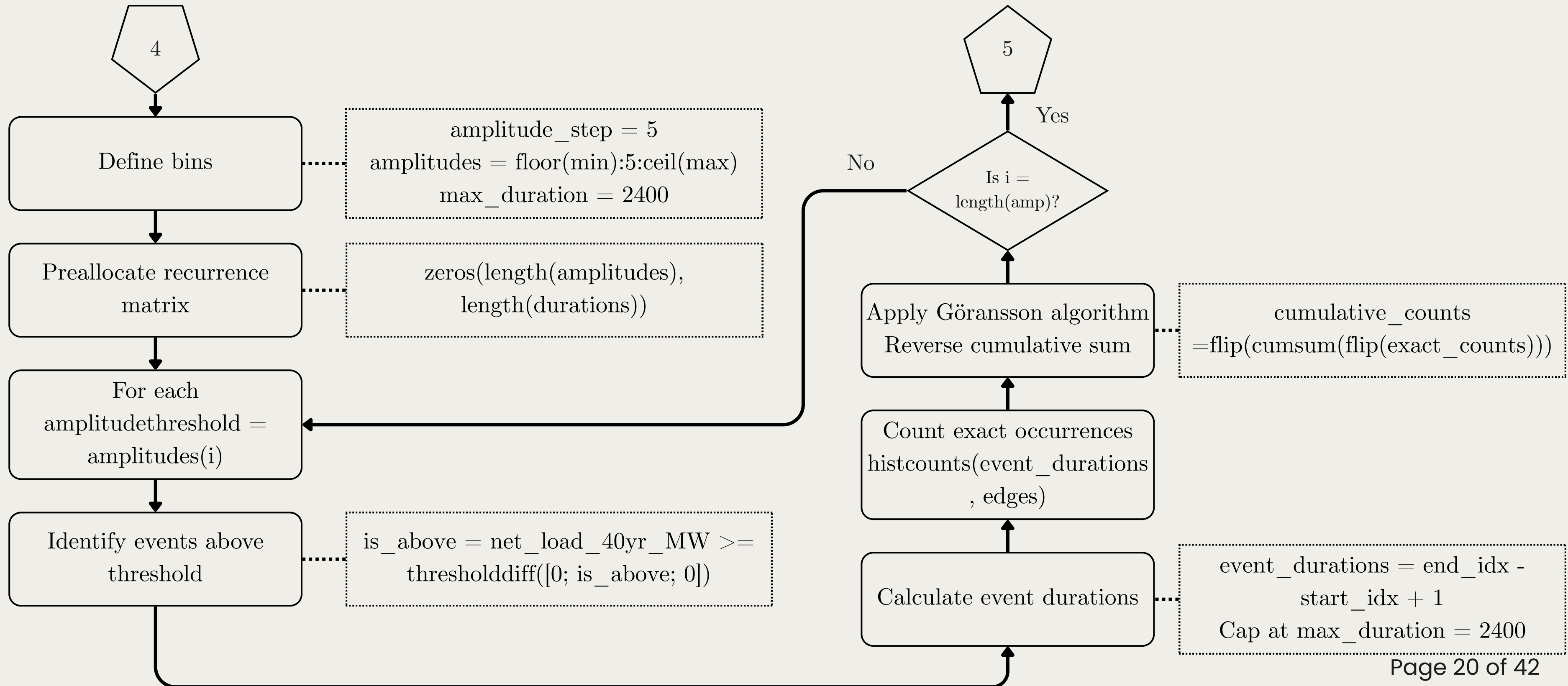
Code Flowchart

UTC to PST and Net Load Calculation



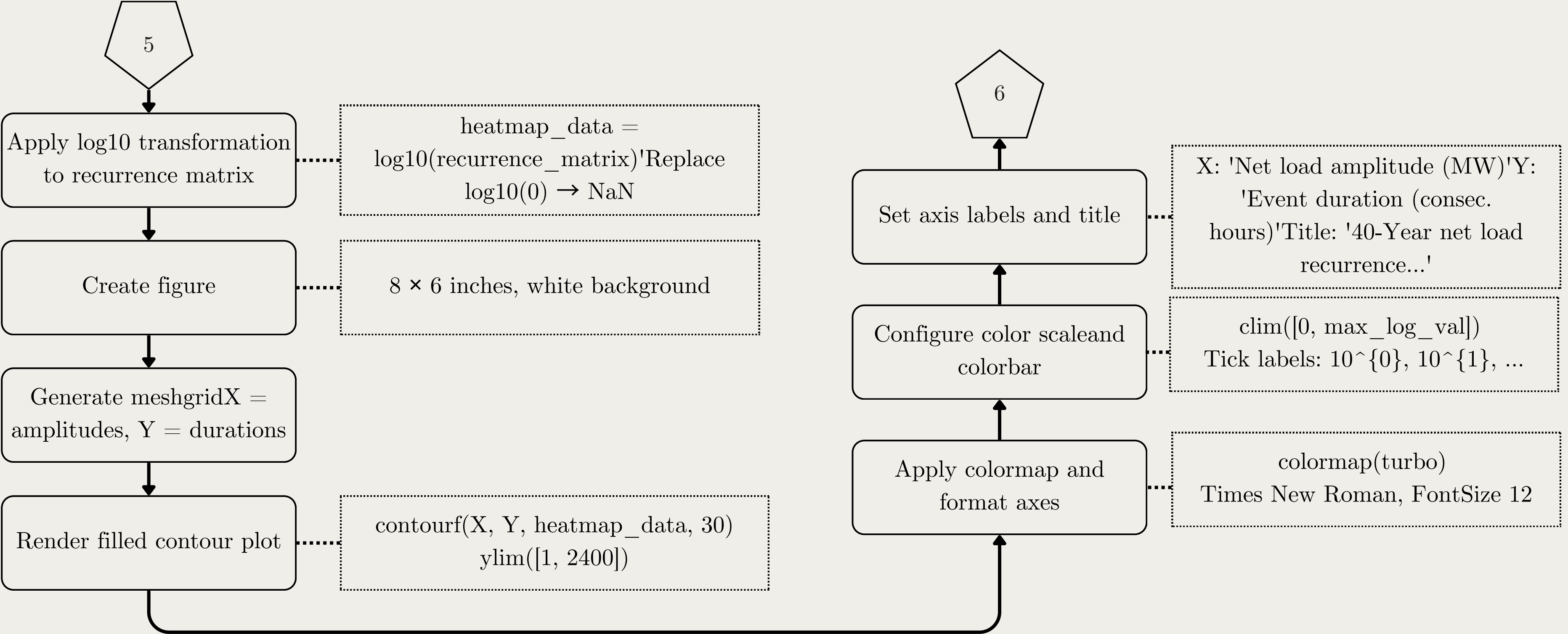
Code Flowchart

Generate Recurrence Matrix (Göransson Continuous Algorithm)



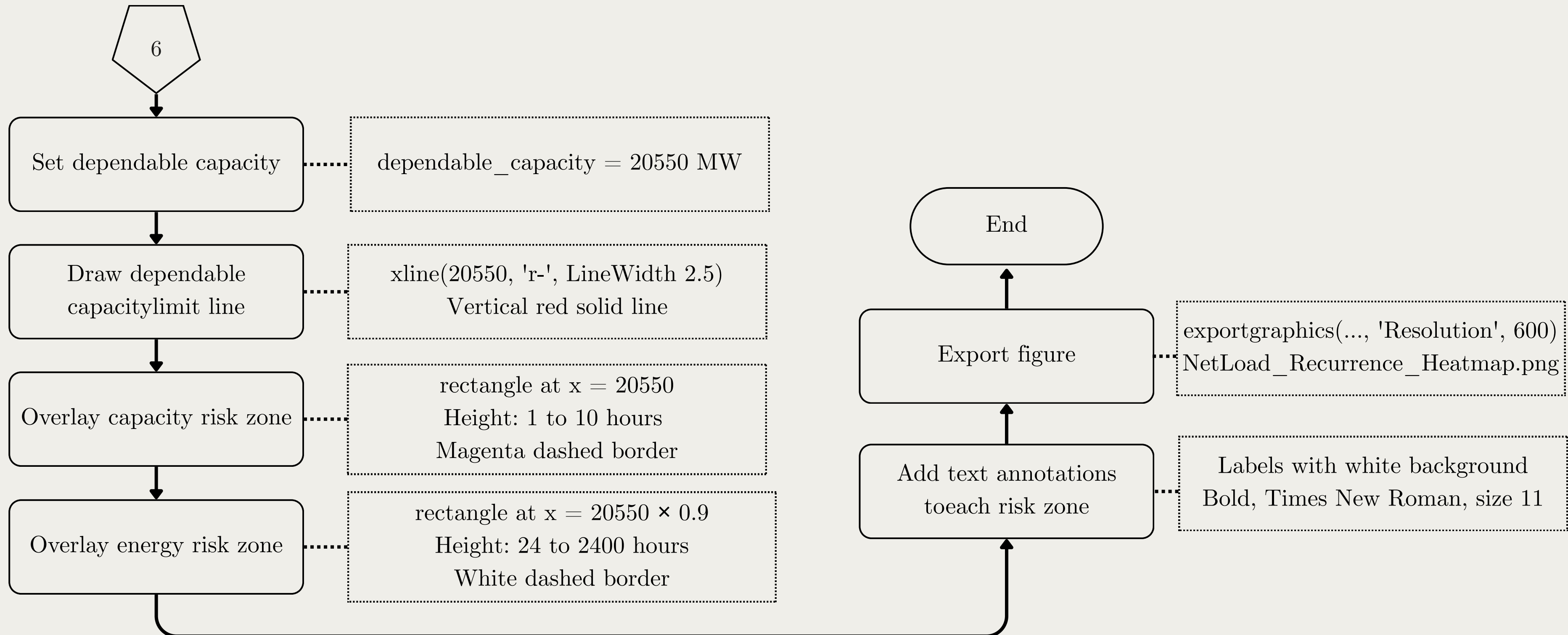
Code Flowchart

Heatmap Plotting



Code Flowchart

Overlay Risk Zones



Demand Profiles

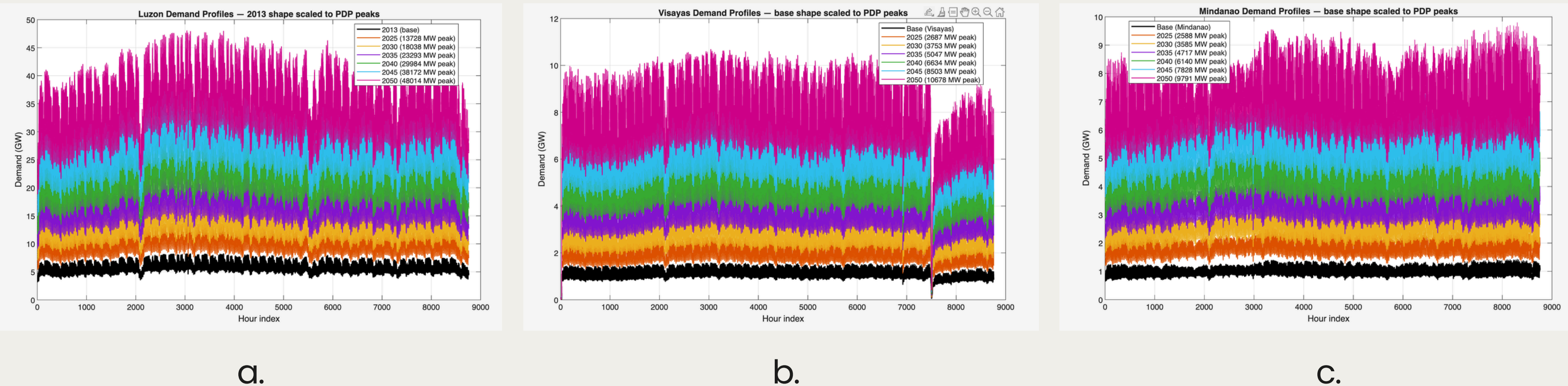


Figure 4: Demand profiles from 2013 hourly demand scaled to PDP years: (a) Luzon, (b) Visayas, (c) Mindanao

Net-load Profiles: Reference Scenario

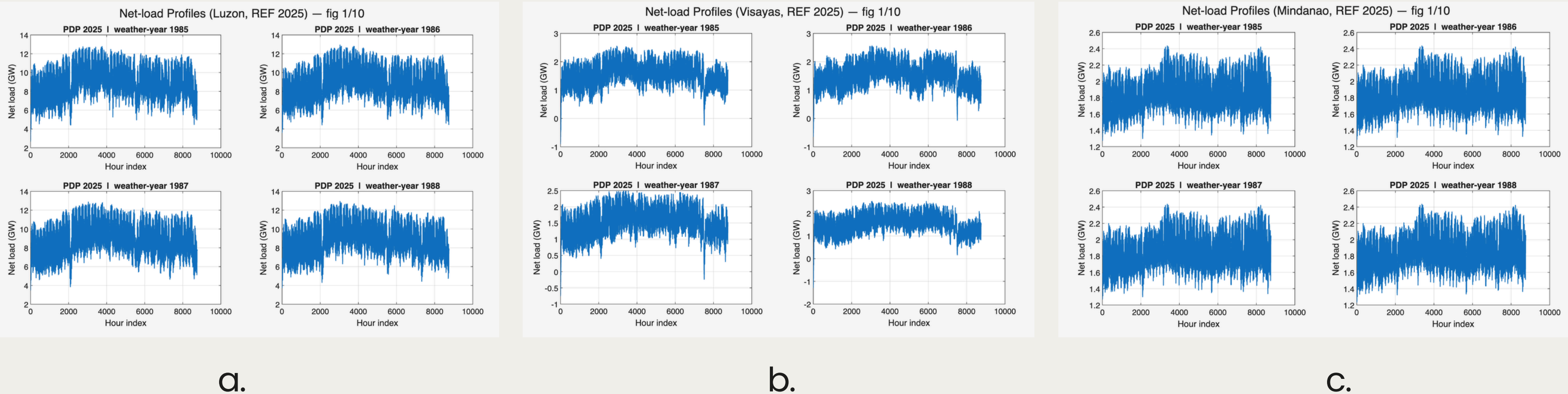
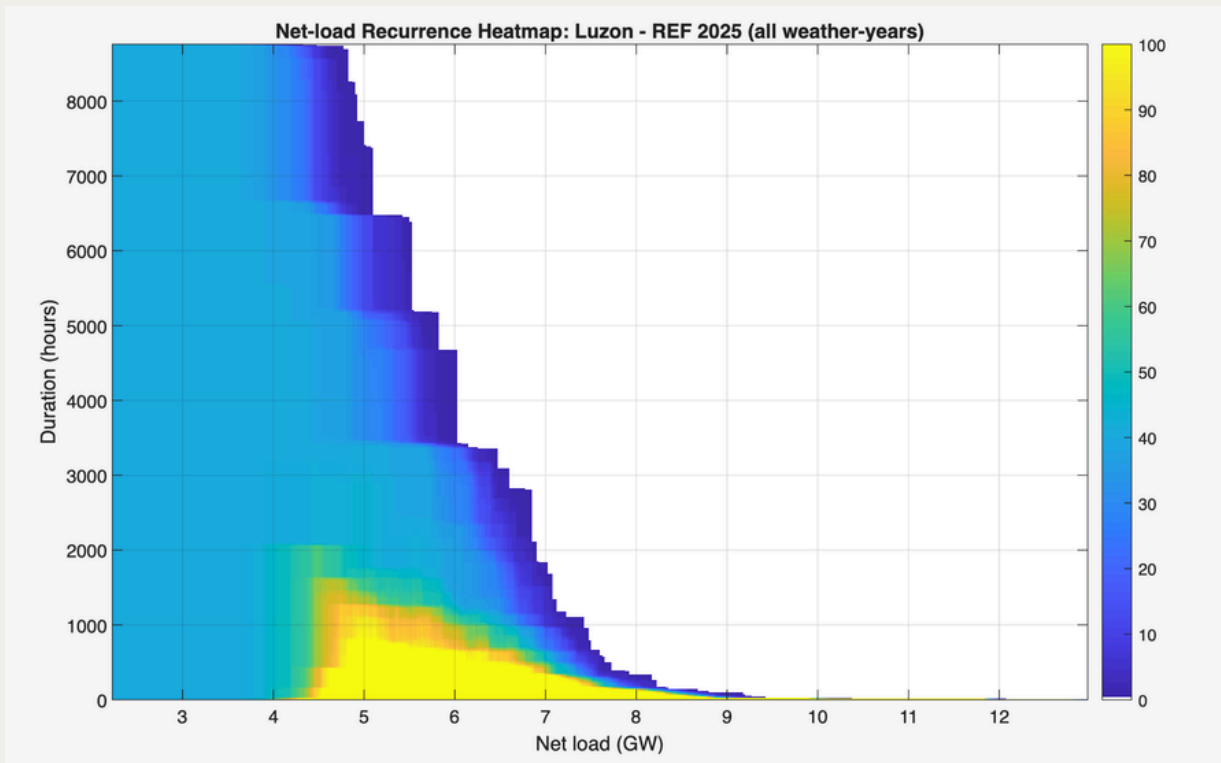


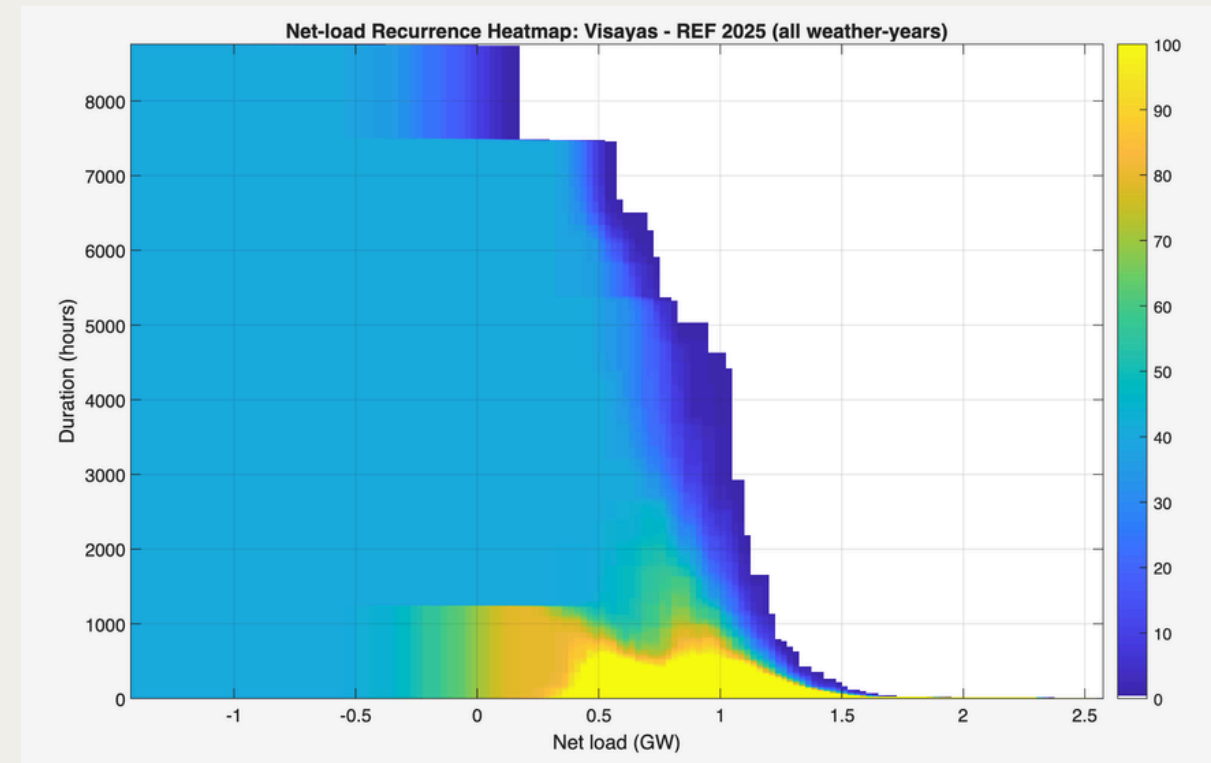
Figure 5: Net-load profiles* for Reference Scenario: (a) Luzon, (b) Visayas, (c) Mindanao

*Only four net-load profiles per grid are presented. There are 40 profiles per grid

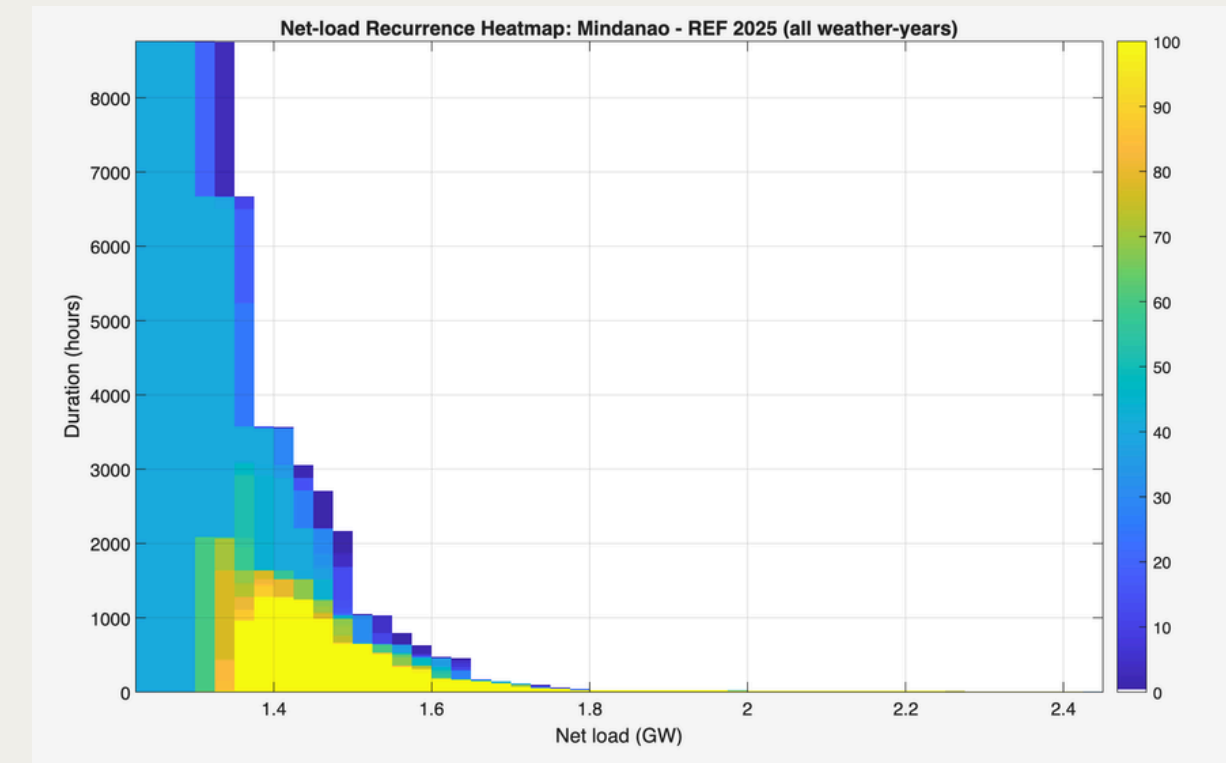
Heatmap: Reference Scenario (2025)



a.



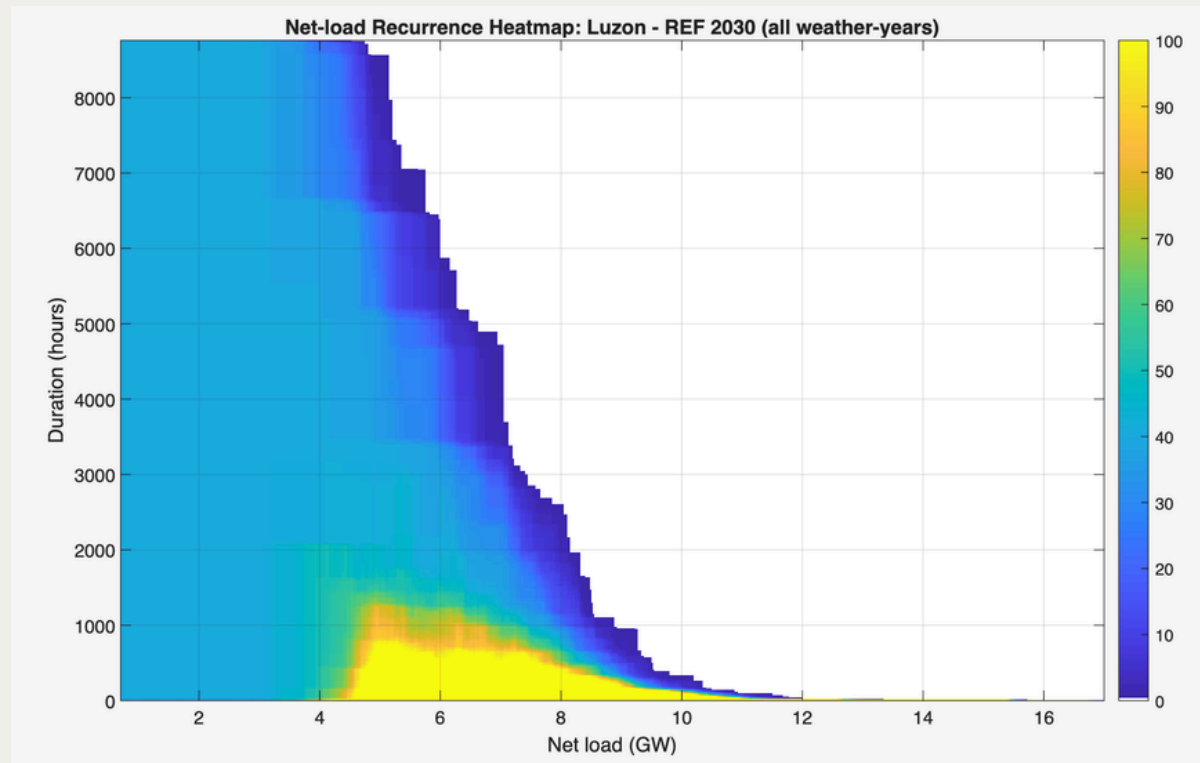
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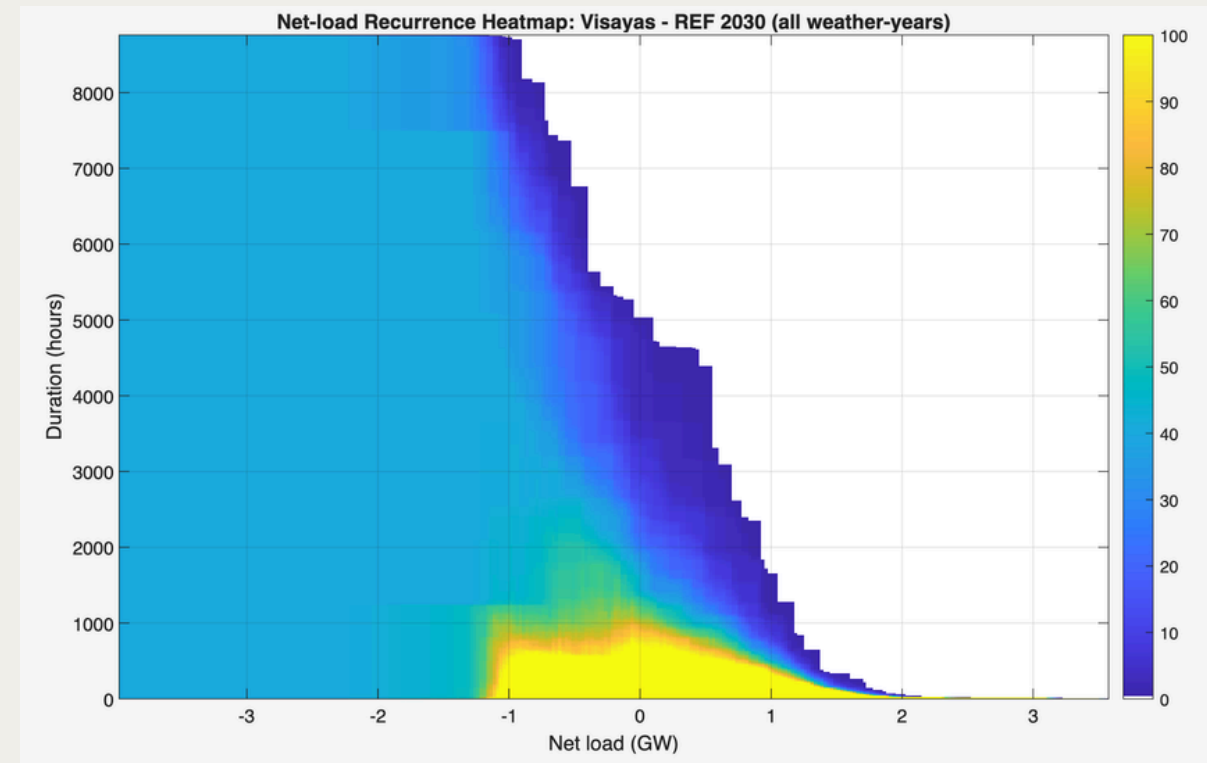
c.

Figure 6: 2025 Reference Scenario Recurrence Heatmap:
(a) Luzon, (b) Visayas, (c) Mindanao

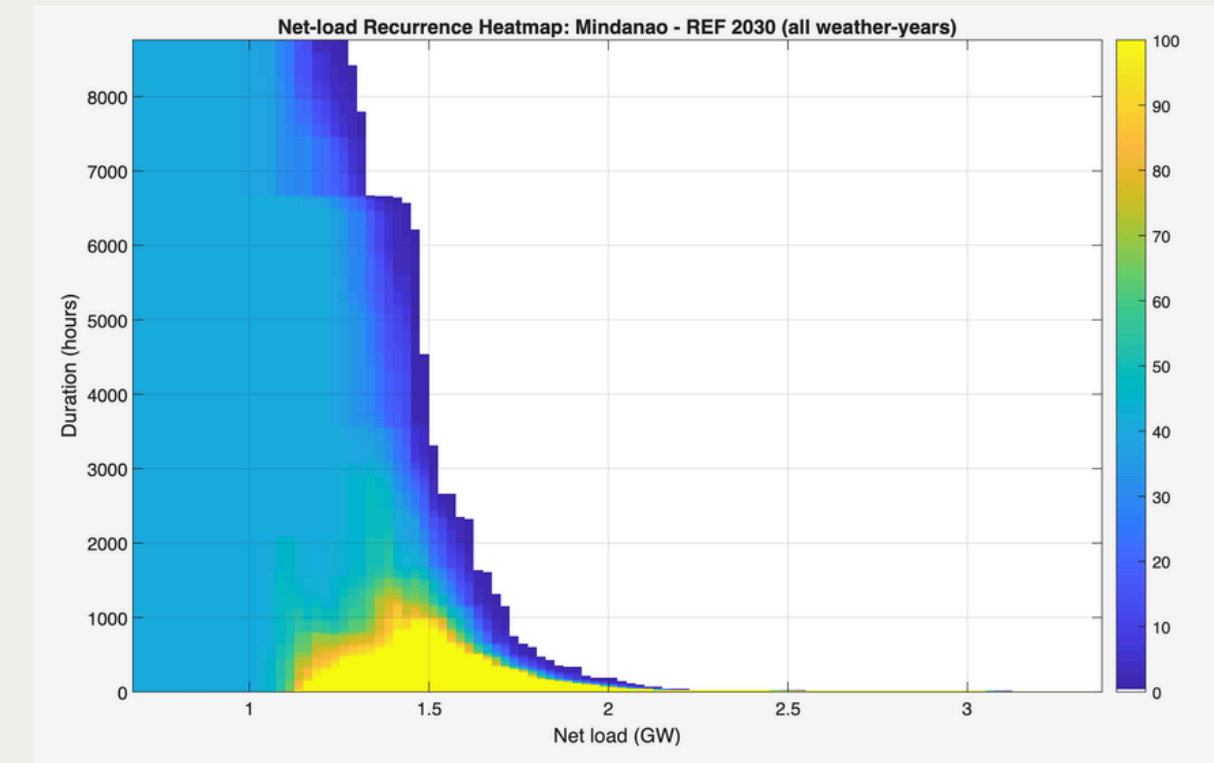
Heatmap: Reference Scenario (2030)



a.



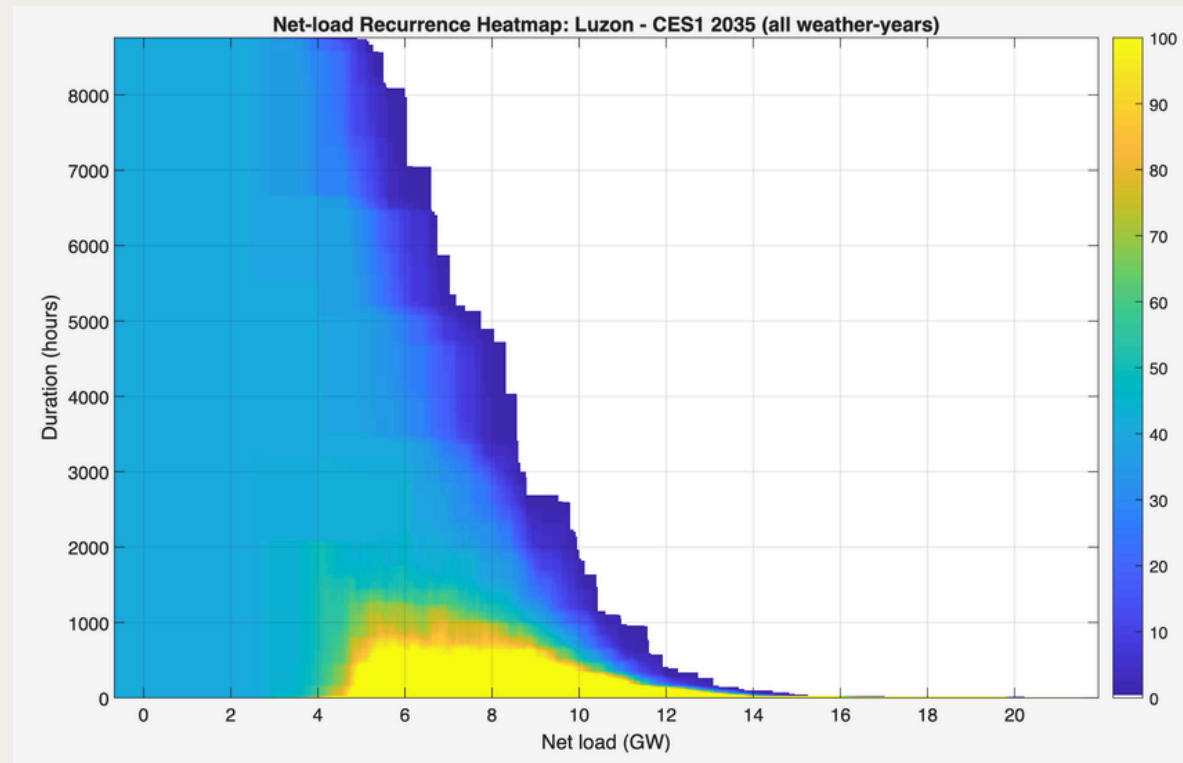
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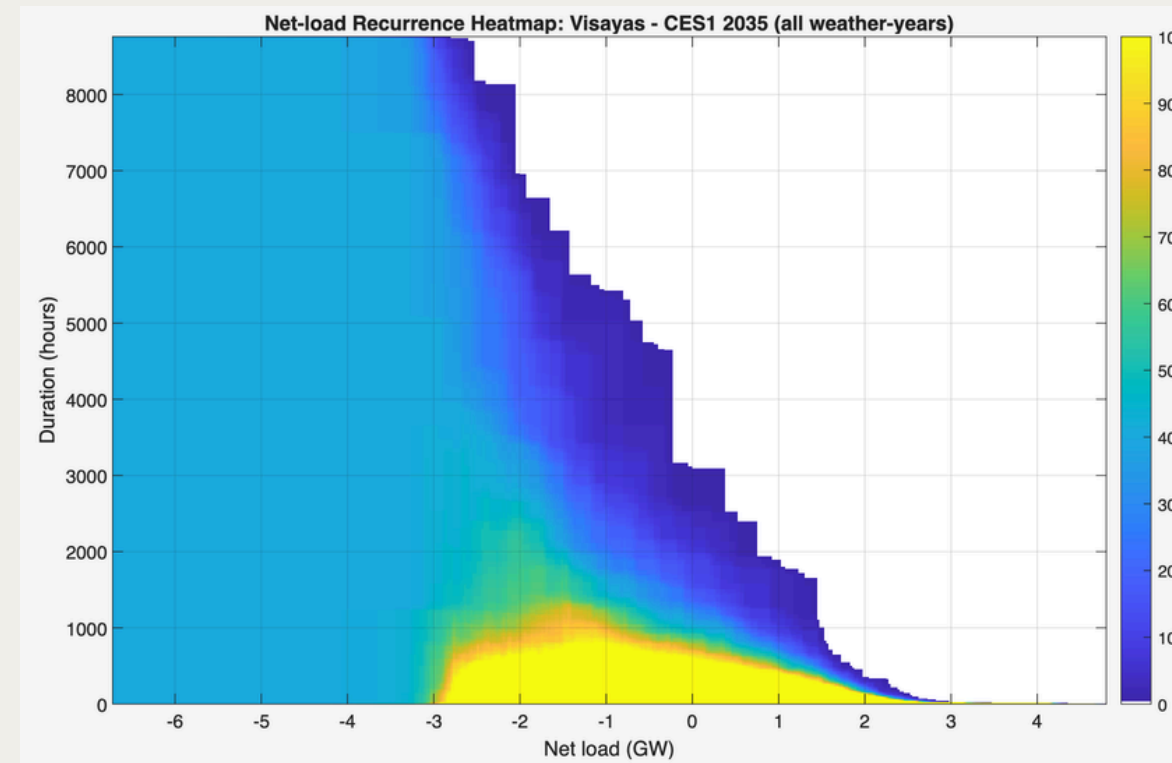
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Figure 7: 2030 Reference Scenario Recurrence Heatmap:
(a) Luzon, (b) Visayas, (c) Mindanao

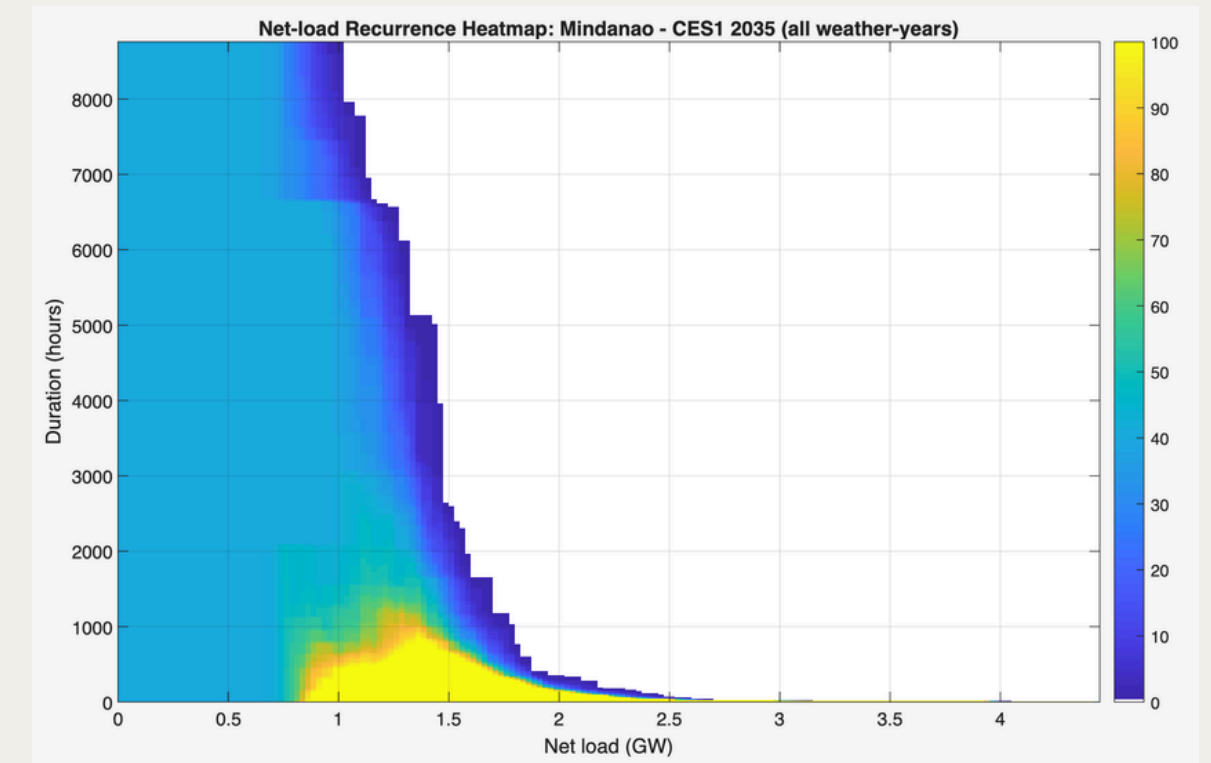
Heatmap: Reference Scenario (2035)



a.



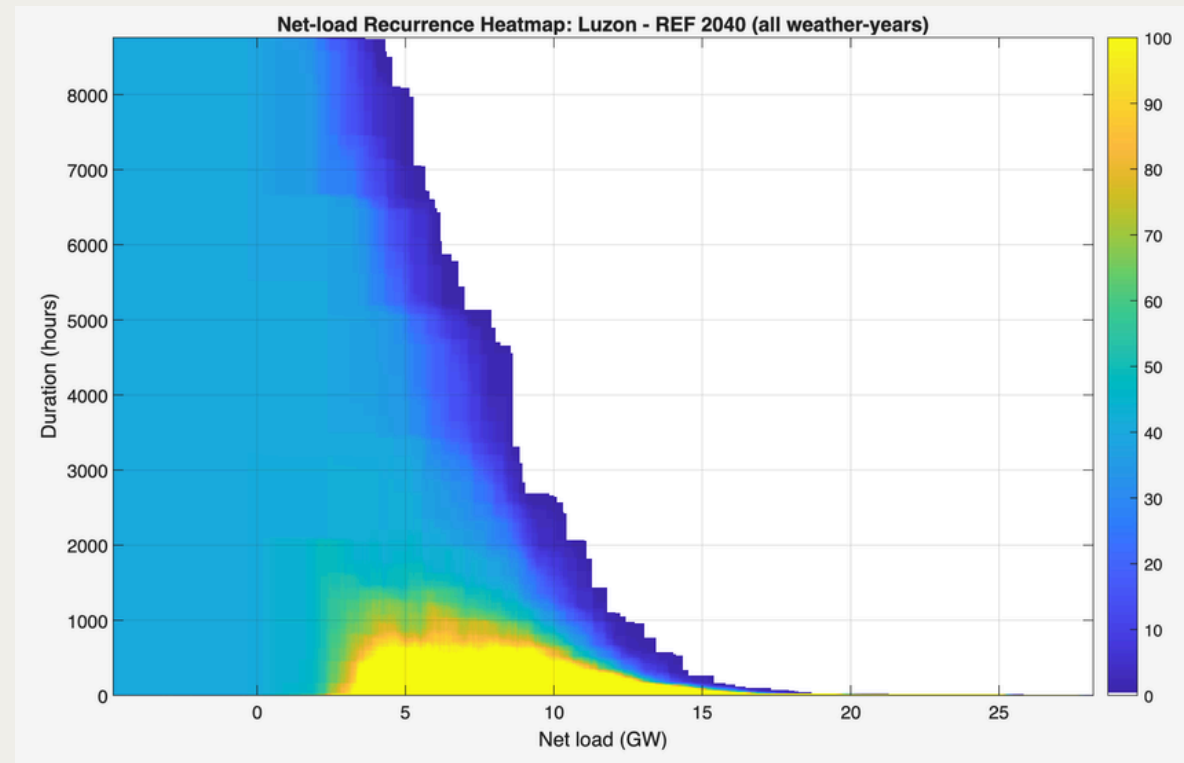
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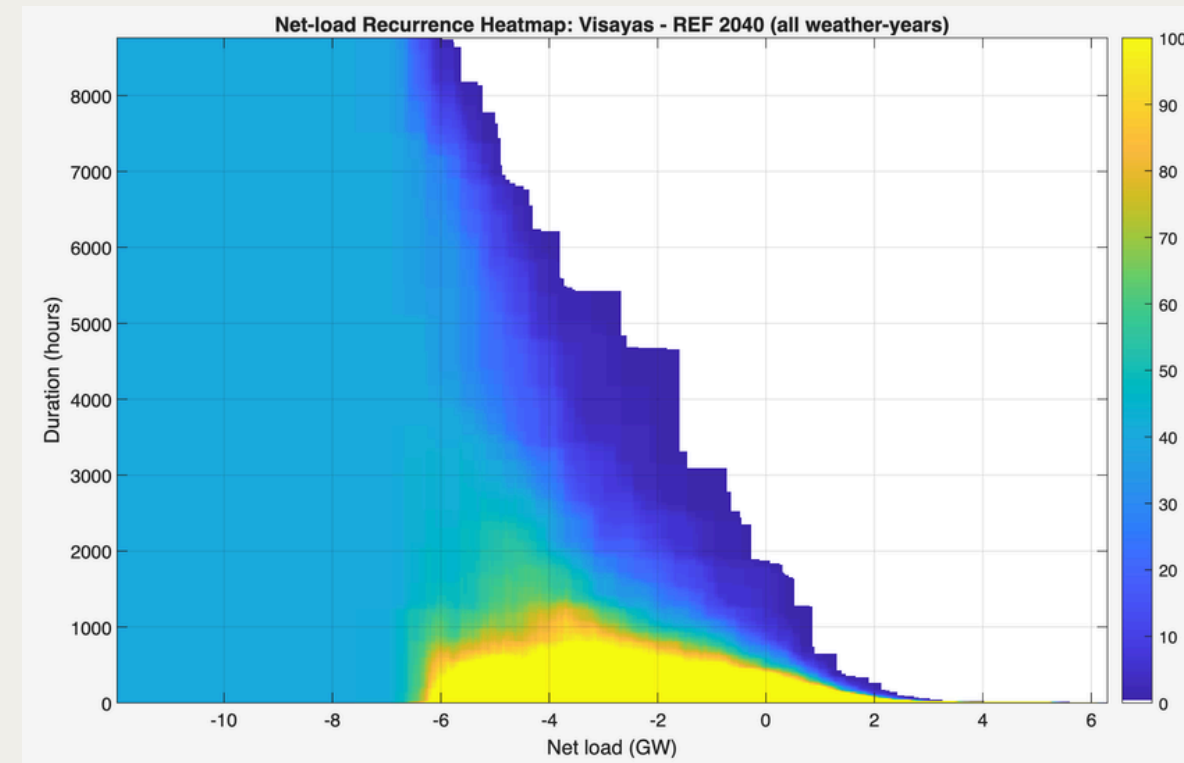
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Figure 8: 2035 Reference Scenario Recurrence Heatmap:
(a) Luzon, (b) Visayas, (c) Mindanao

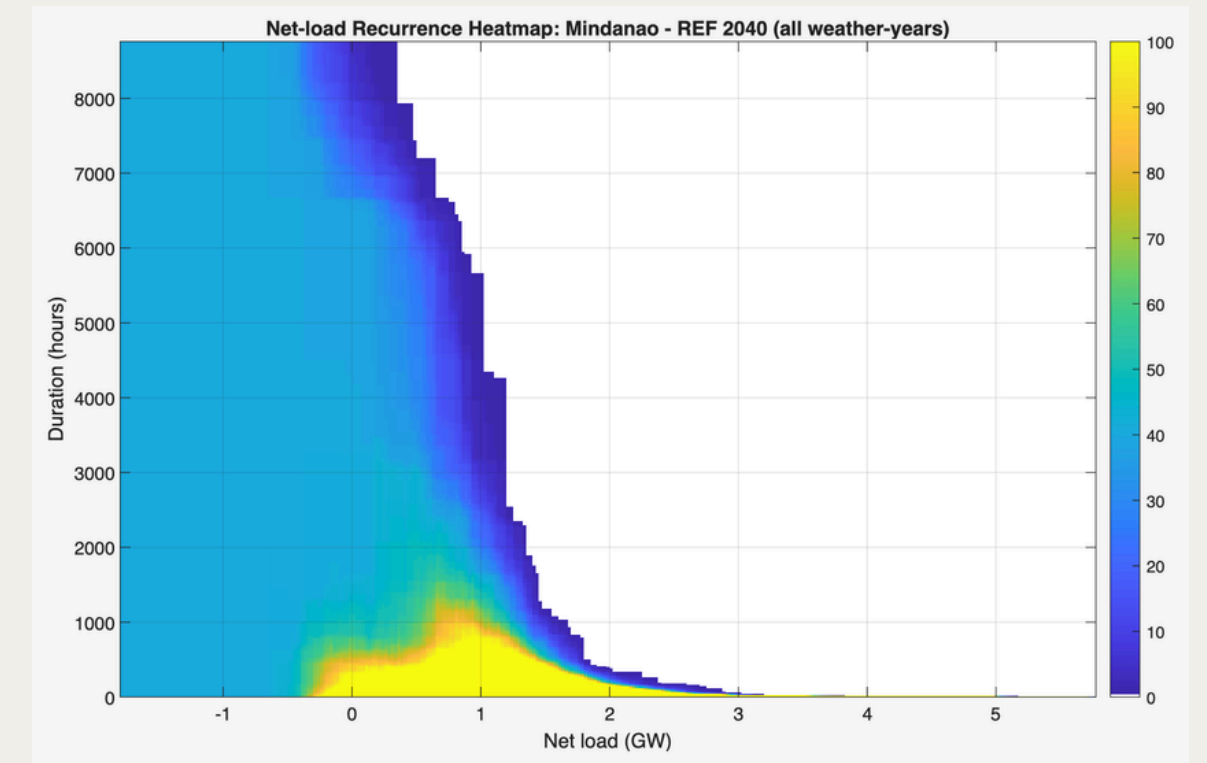
Heatmap: Reference Scenario (2040)



a.



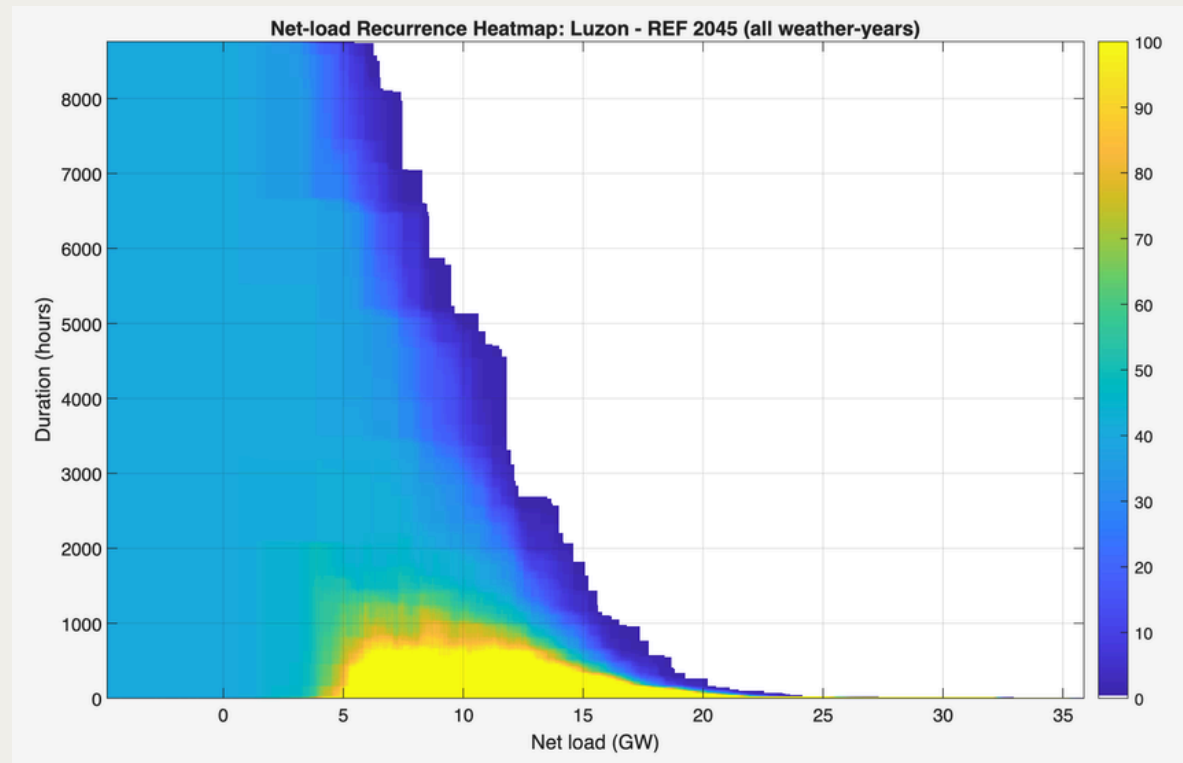
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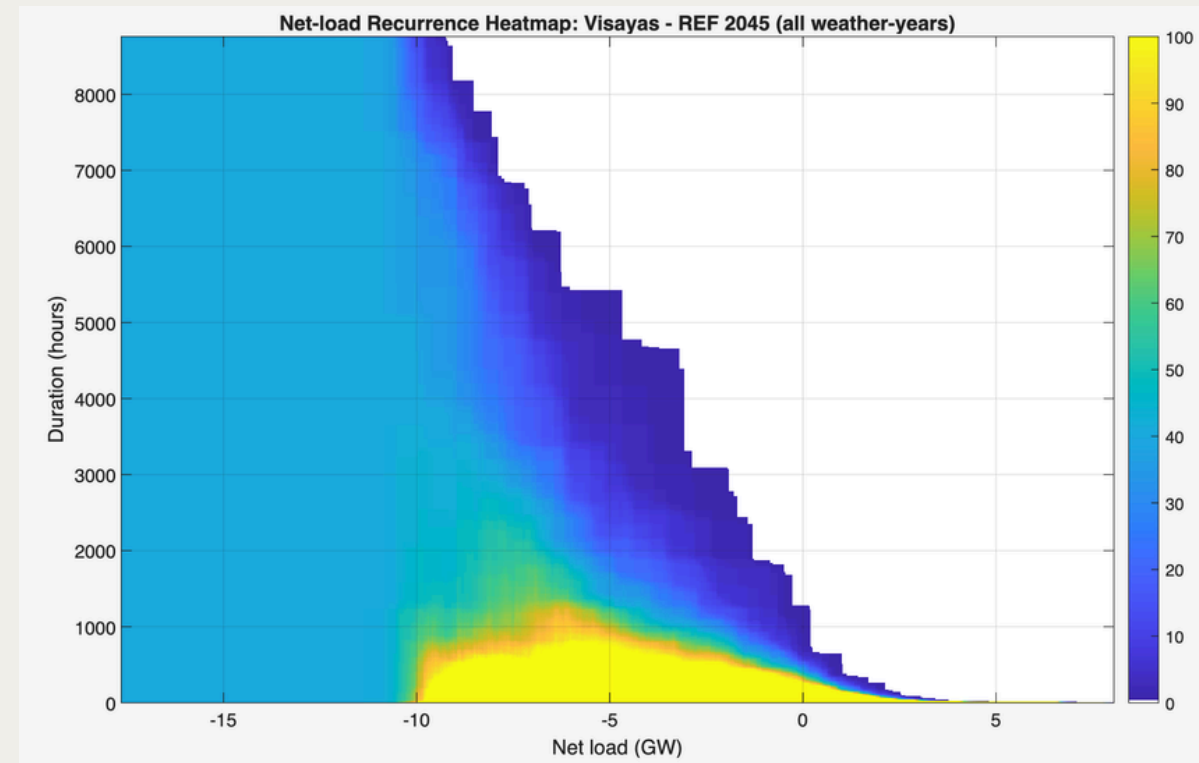
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Figure 9: 2040 Reference Scenario Recurrence Heatmap:
(a) Luzon, (b) Visayas, (c) Mindanao

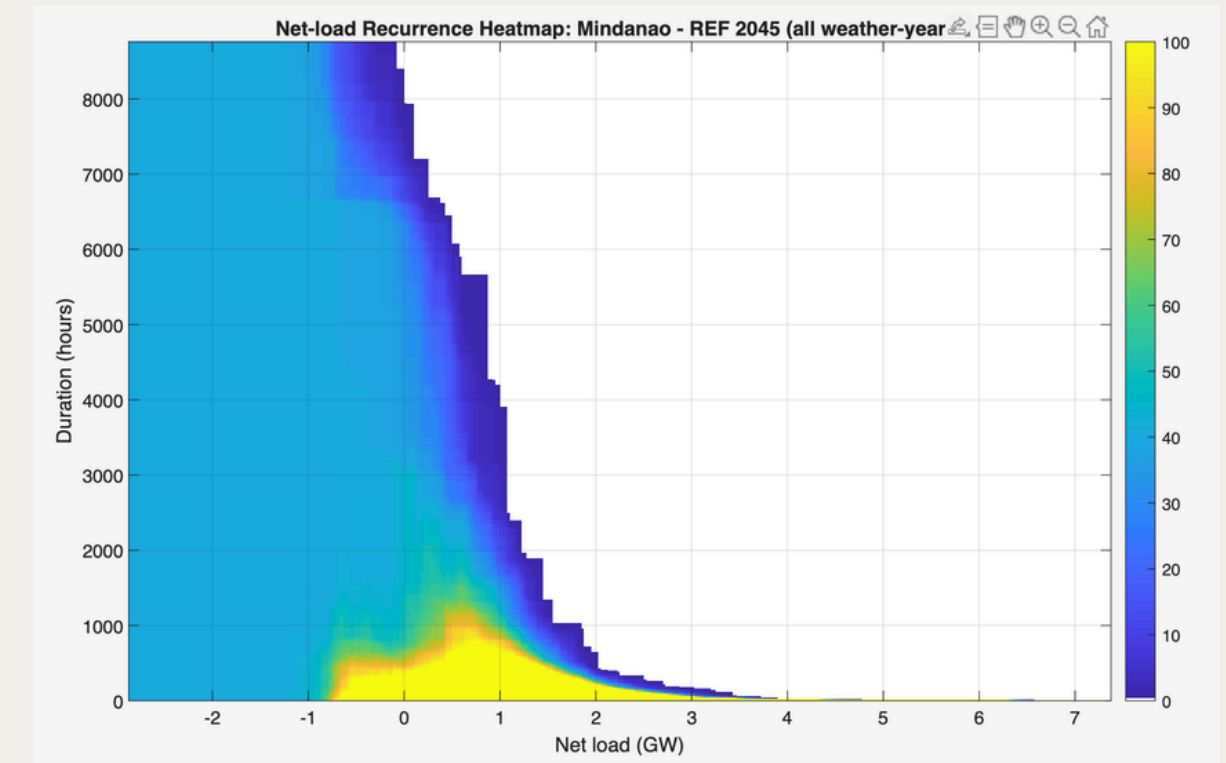
Heatmap: Reference Scenario (2045)



a.



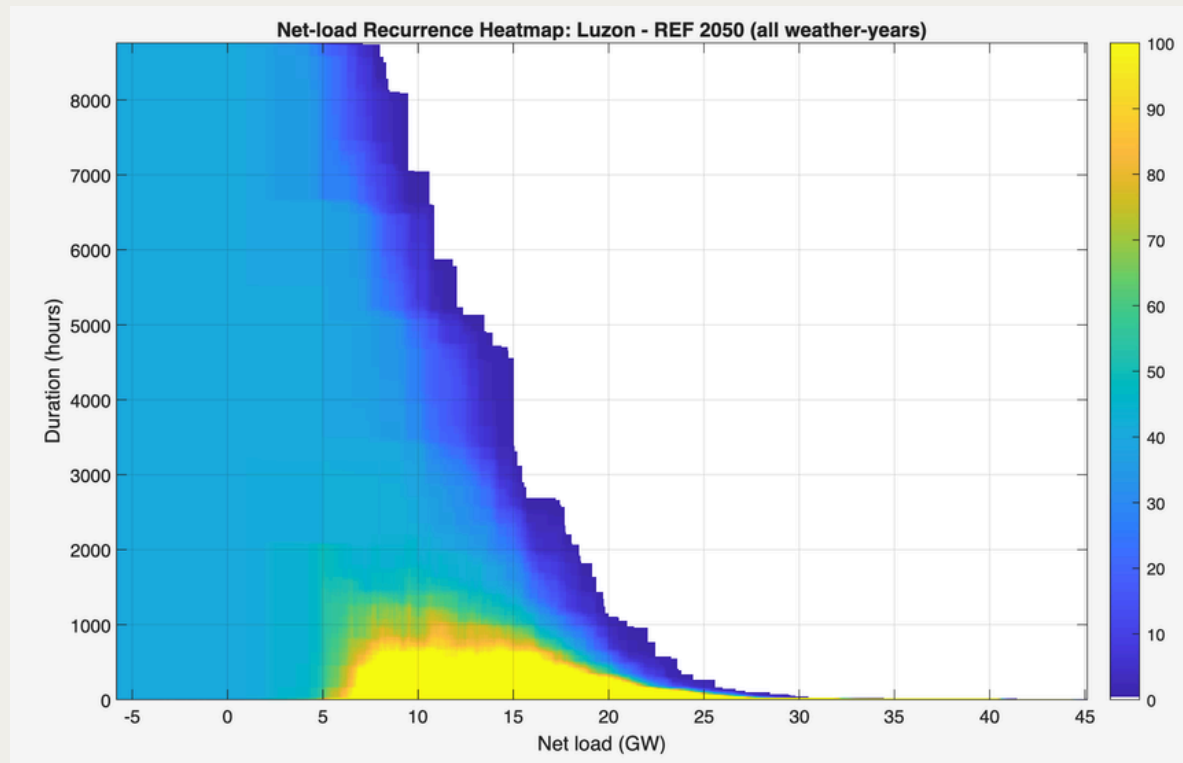
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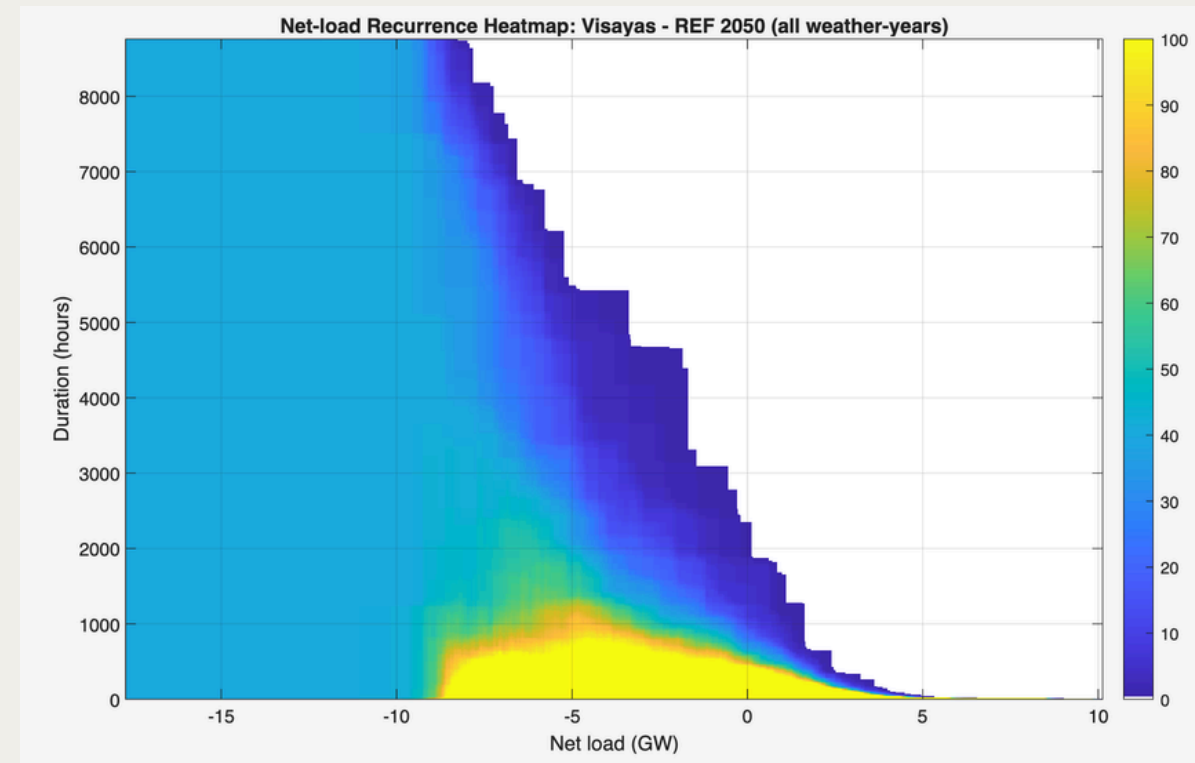
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Figure 10: 2045 Reference Scenario Recurrence Heatmap:
(a) Luzon, (b) Visayas, (c) Mindanao

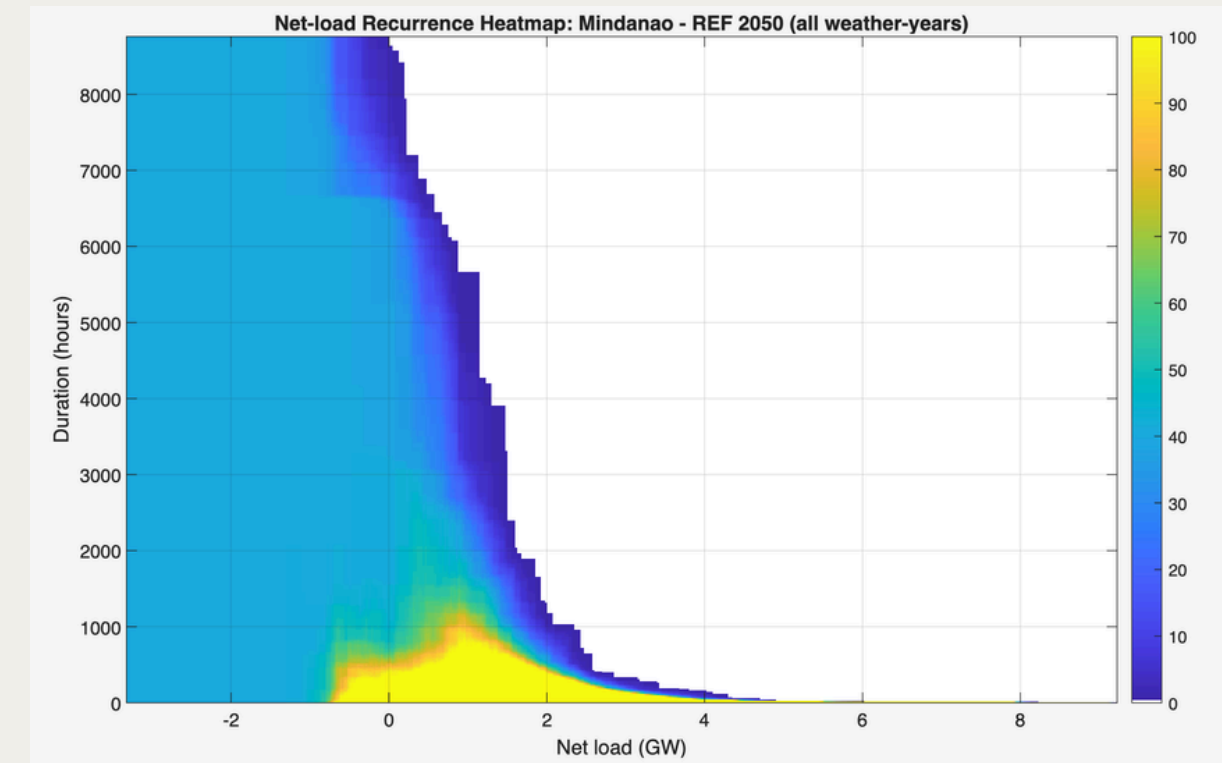
Heatmap: Reference Scenario (2050)



a.



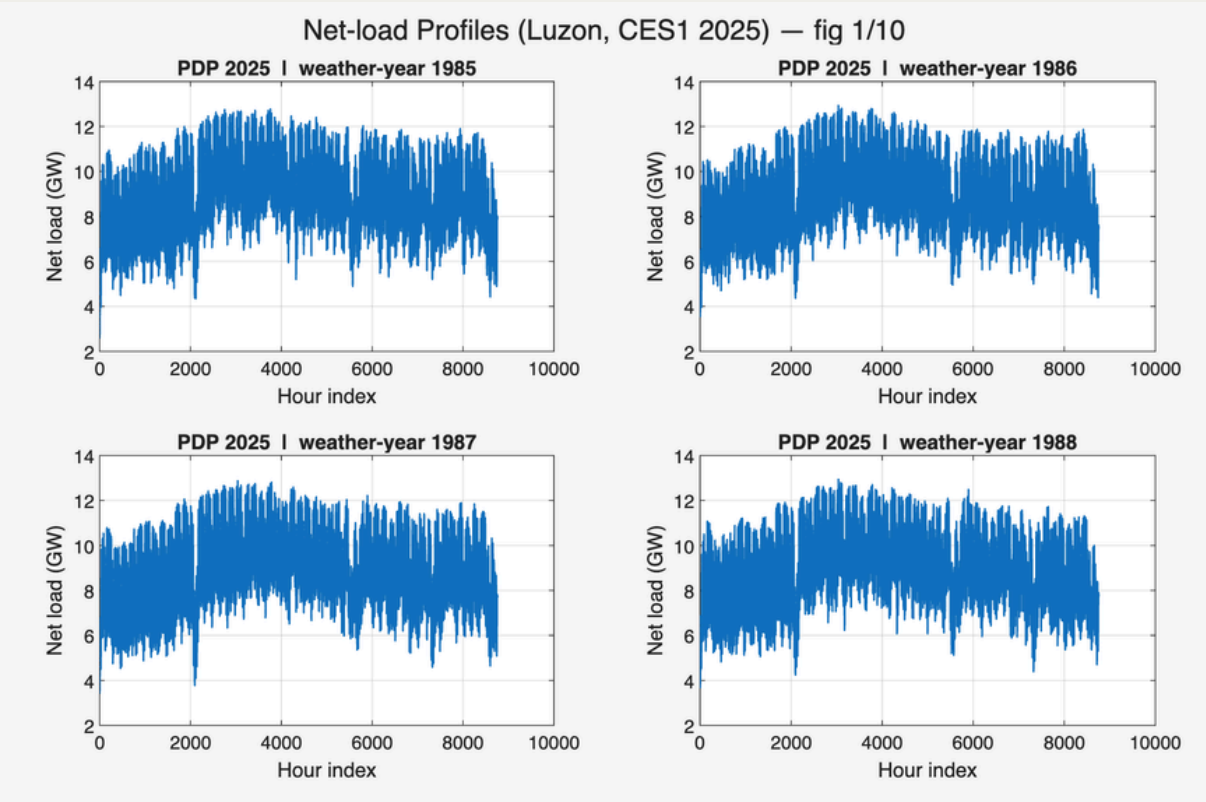
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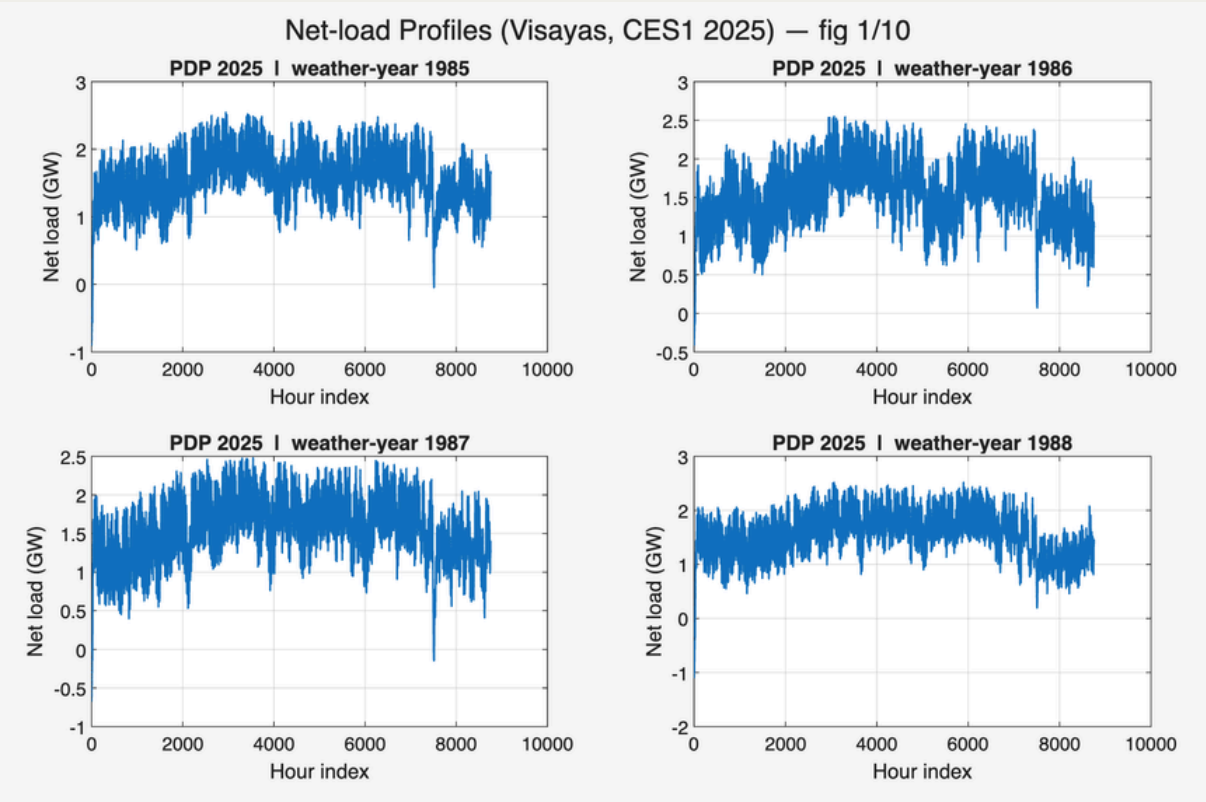
c.

Figure 11: 2050 Reference Scenario Recurrence Heatmap:
(a) Luzon, (b) Visayas, (c) Mindanao

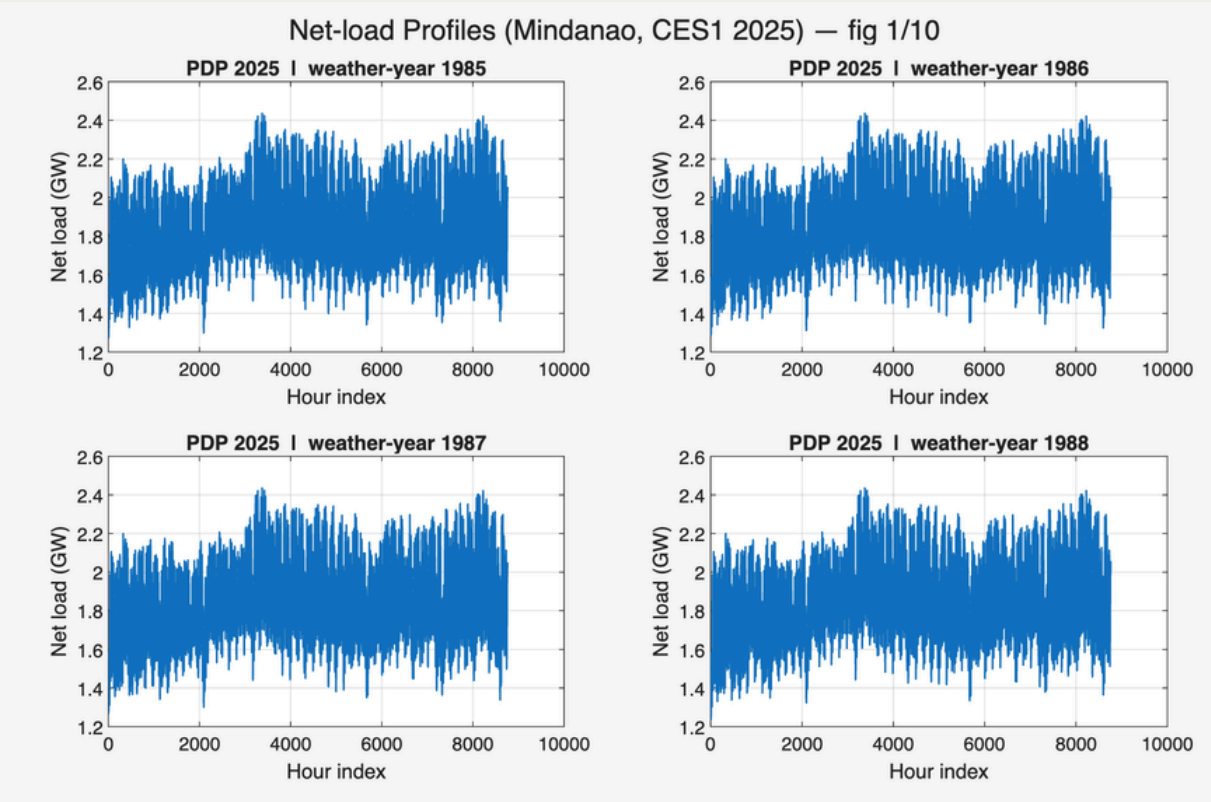
Net-load Profiles: CES 1



a.



b.

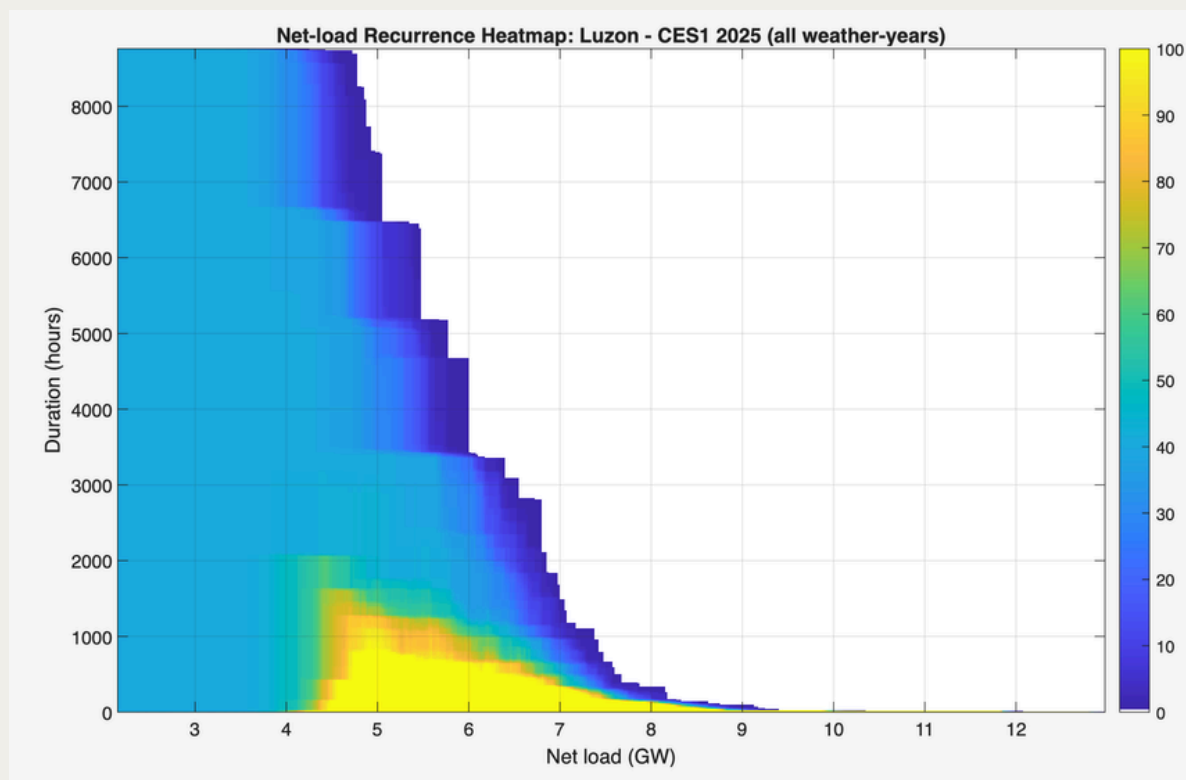


c.

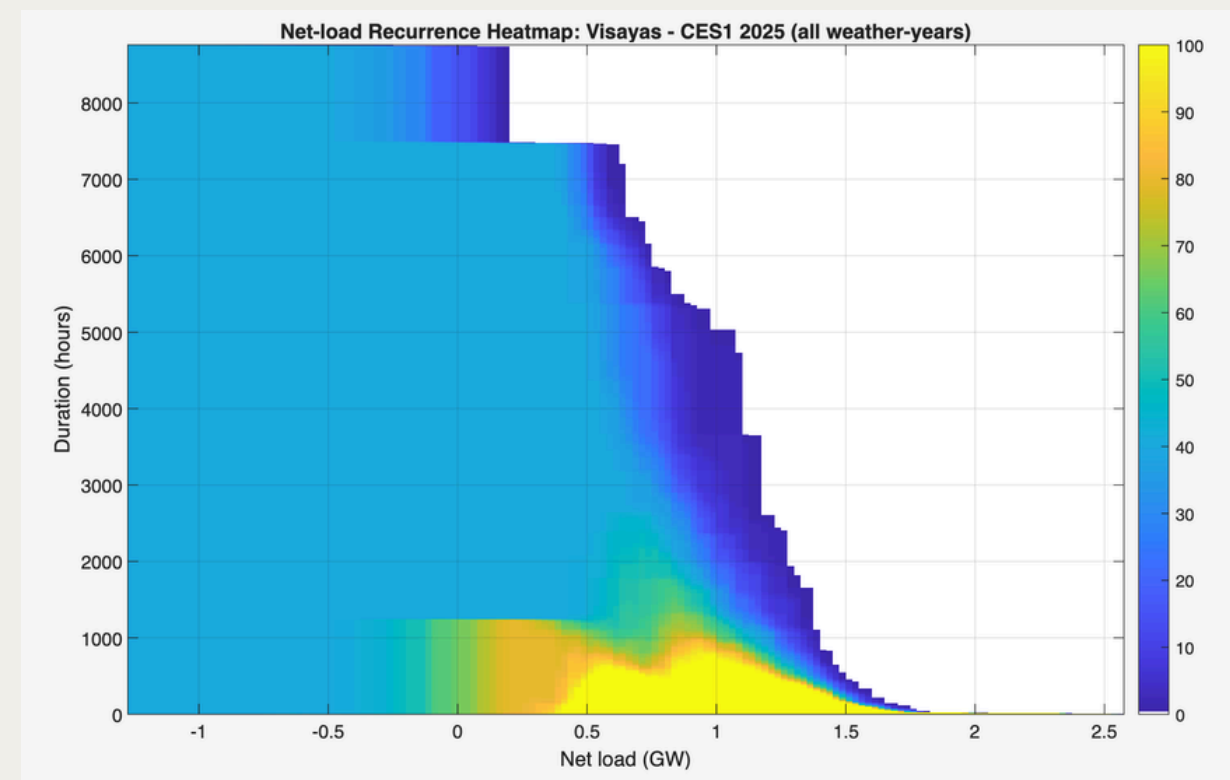
Figure 5: Net-load profiles* for CES 1: (a) Luzon, (b) Visayas, (c) Mindanao

*Only four net-load profiles per grid are presented. There are 40 profiles per grid

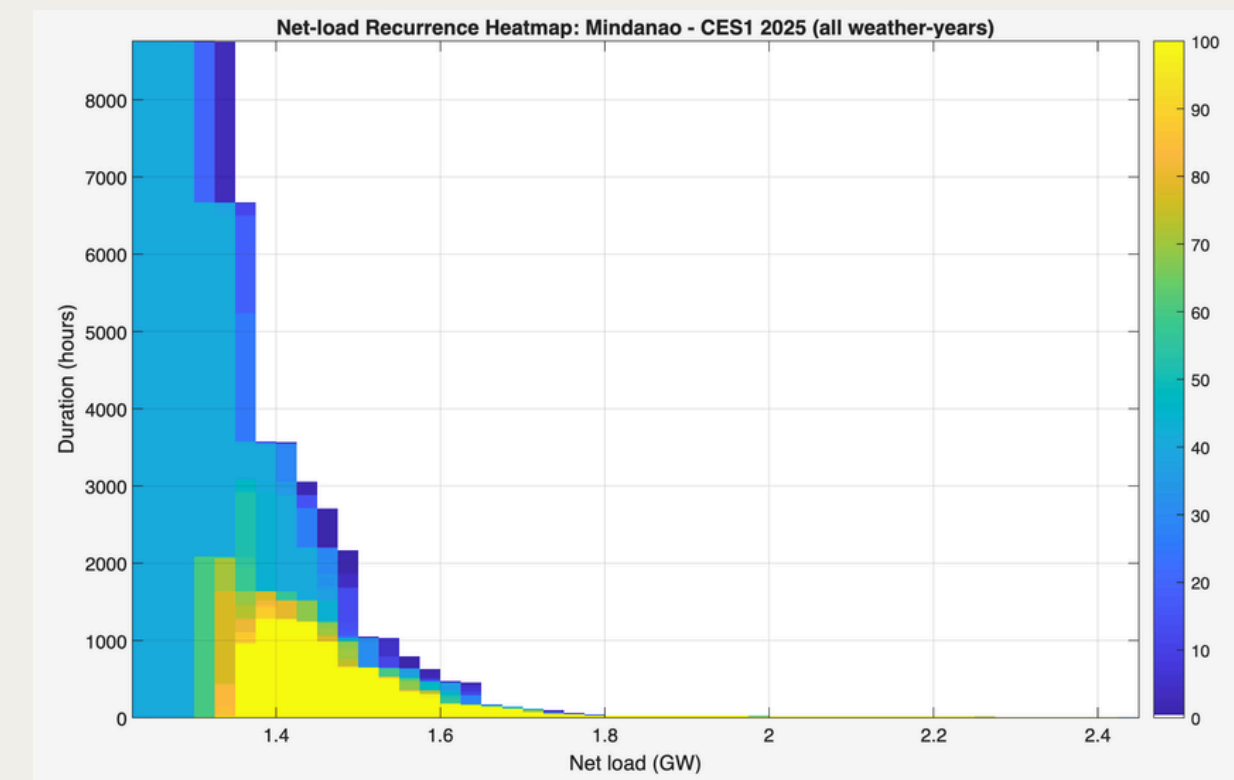
Heatmap: CES 1 (2025)



a.



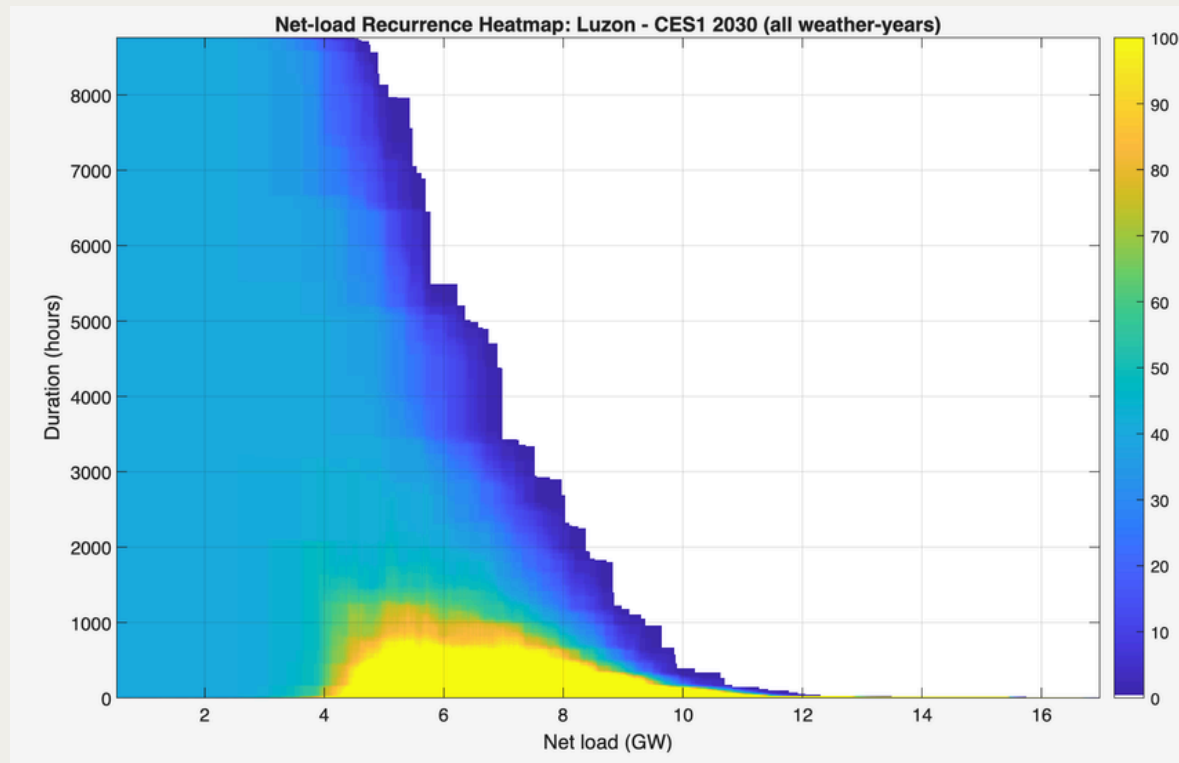
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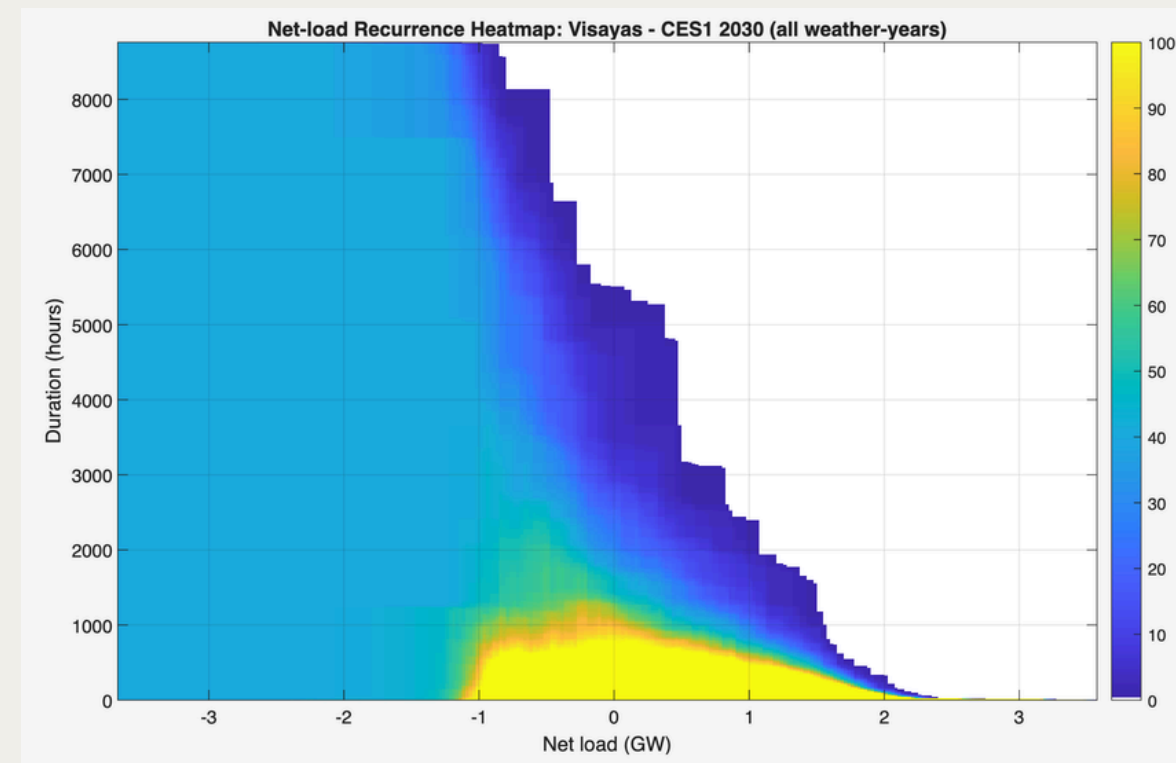
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Figure 6: 2025 CES 1 Recurrence Heatmap:
(a) Luzon, (b) Visayas, (c) Mindanao

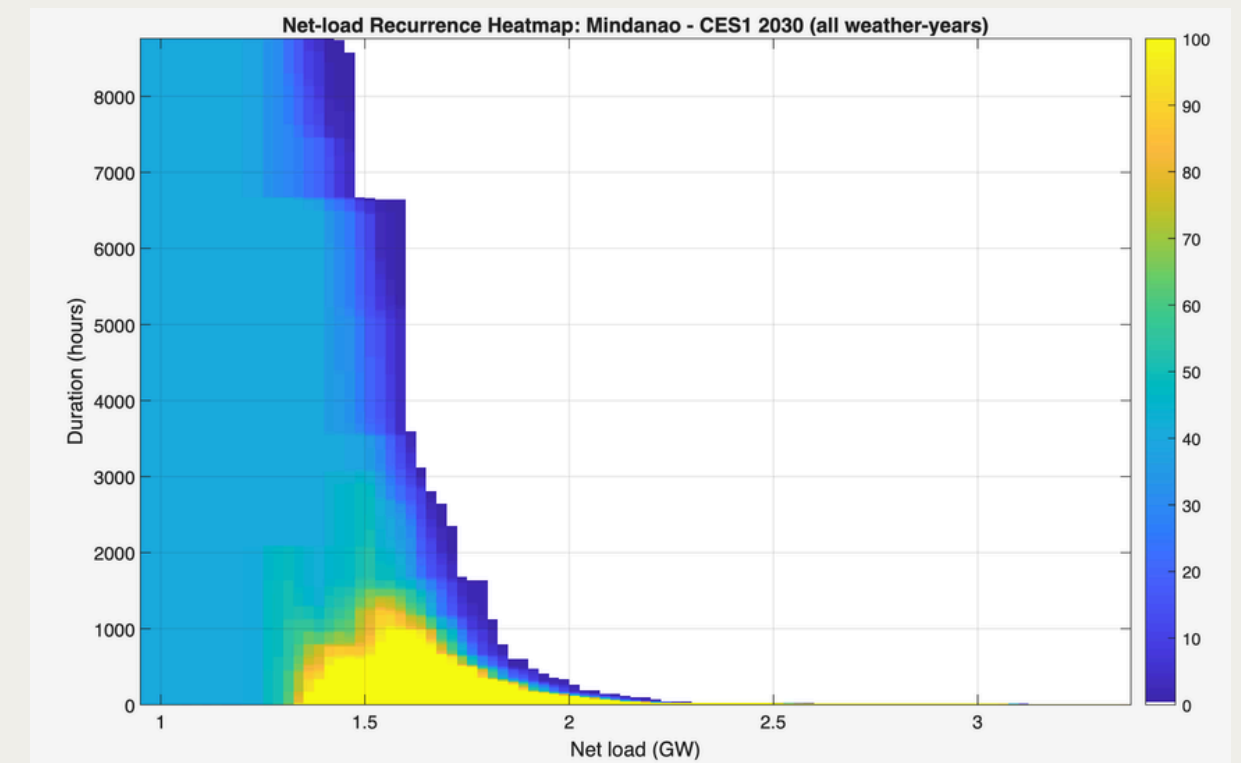
Heatmap: CES 1 (2030)



a.



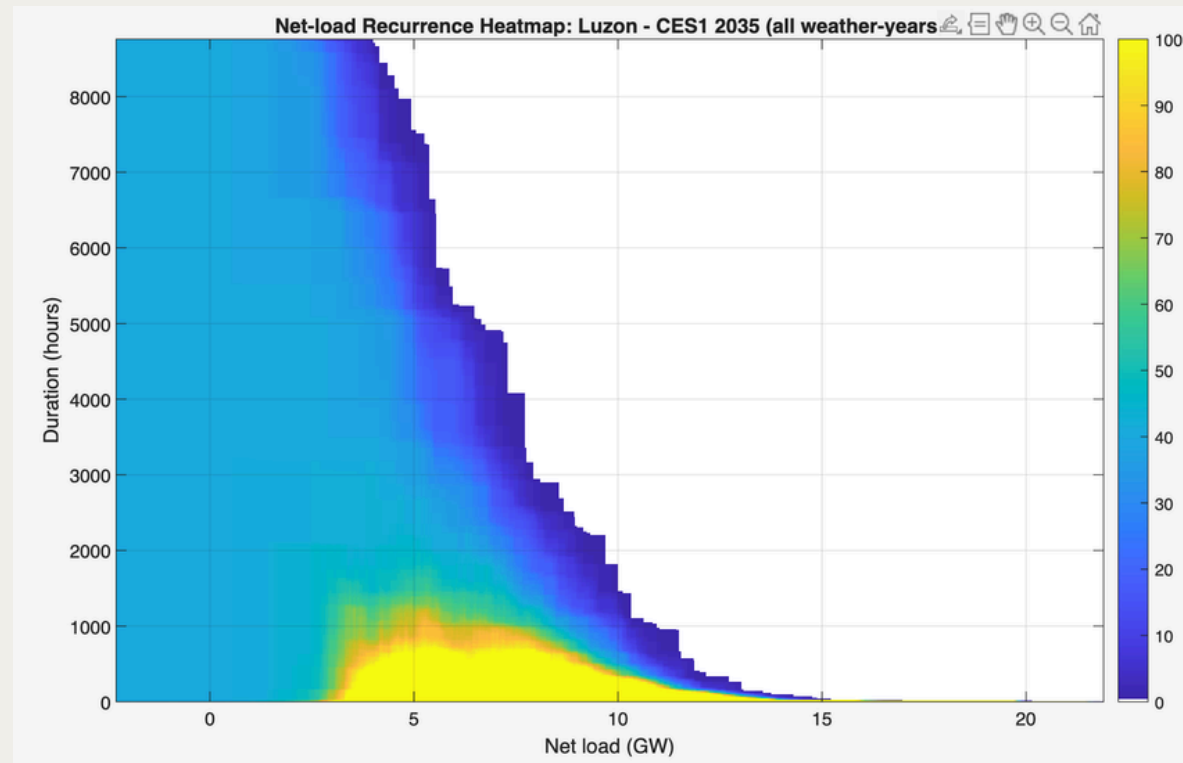
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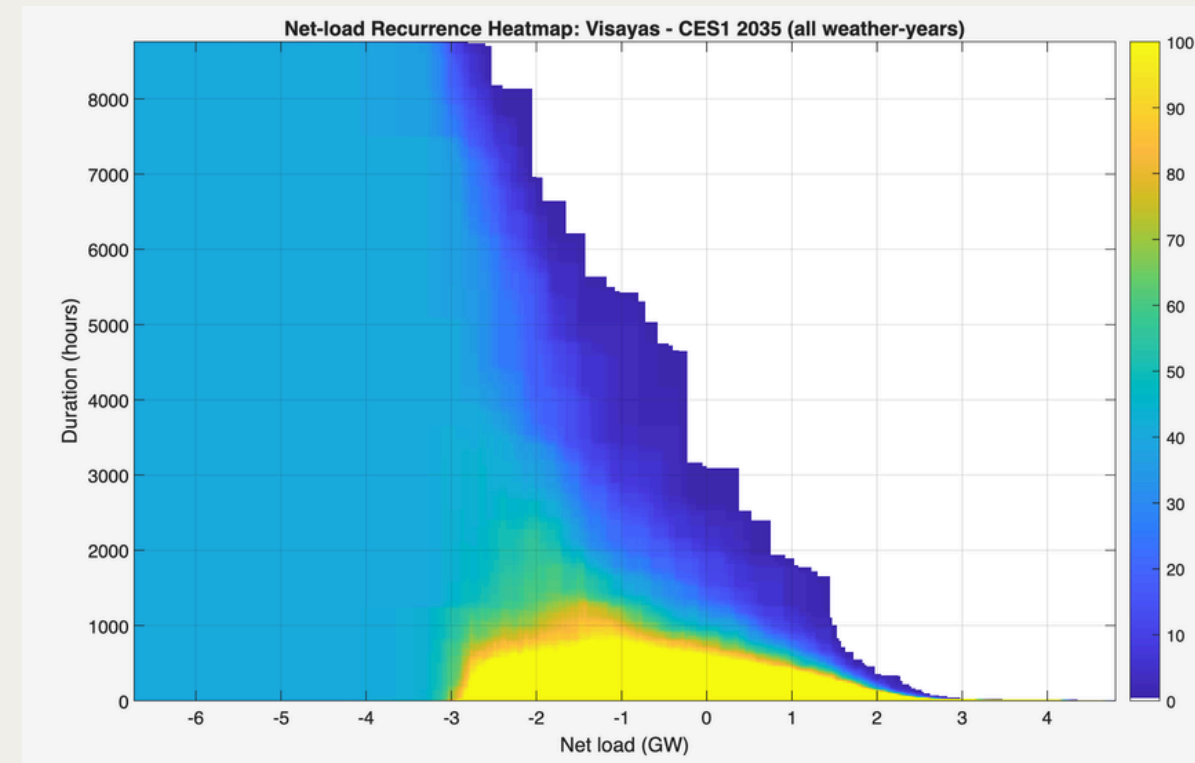
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Figure 7: 2030 CES 1 Recurrence Heatmap:
(a) Luzon, (b) Visayas, (c) Mindanao

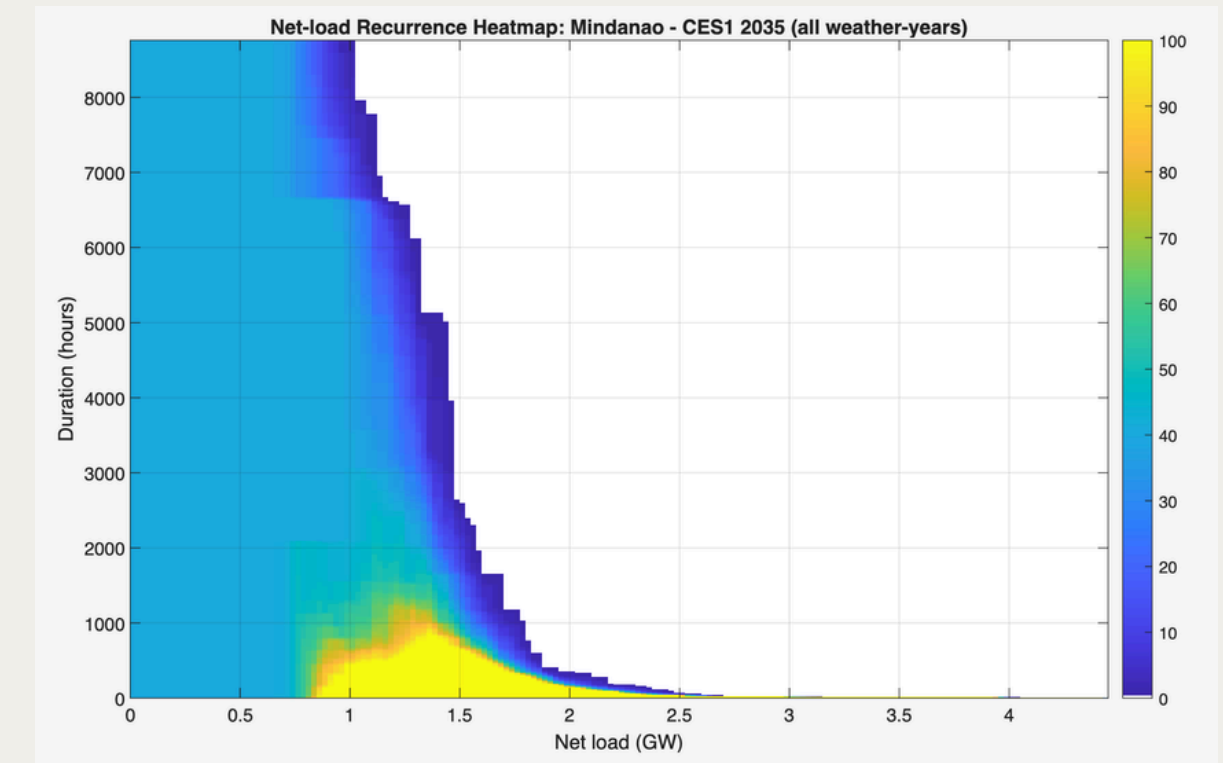
Heatmap: CES 1 (2035)



a.



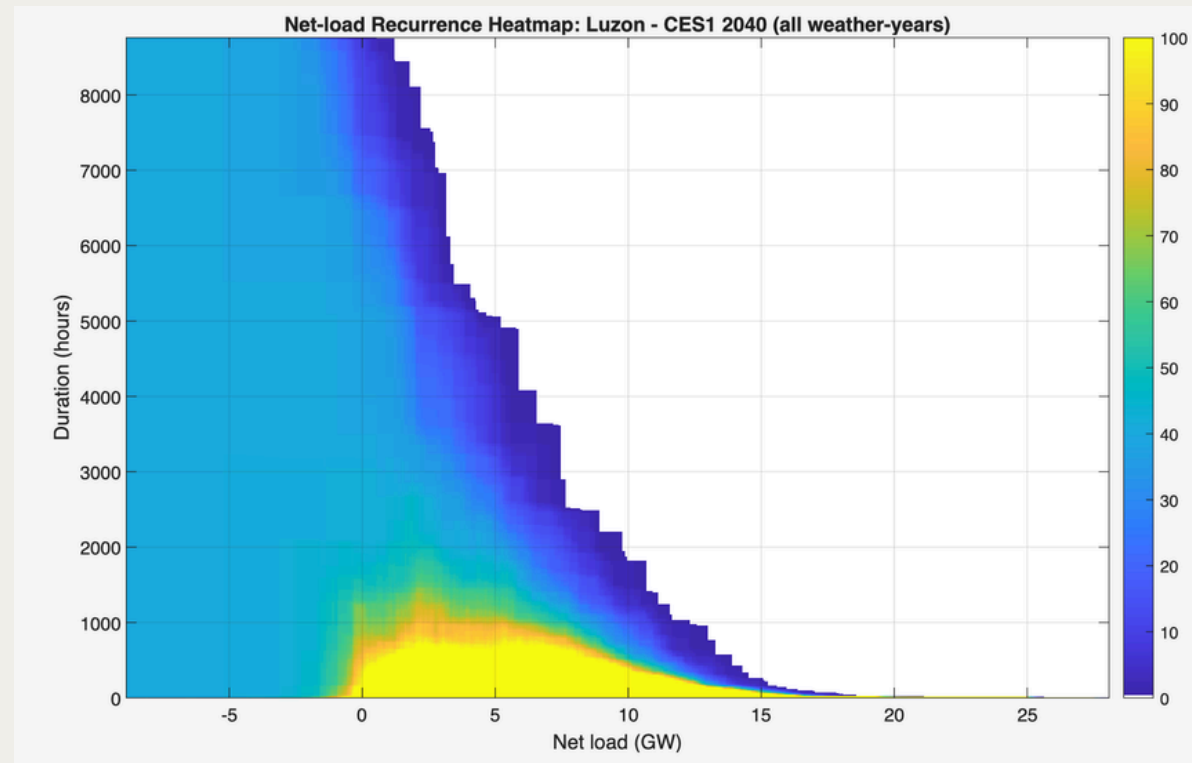
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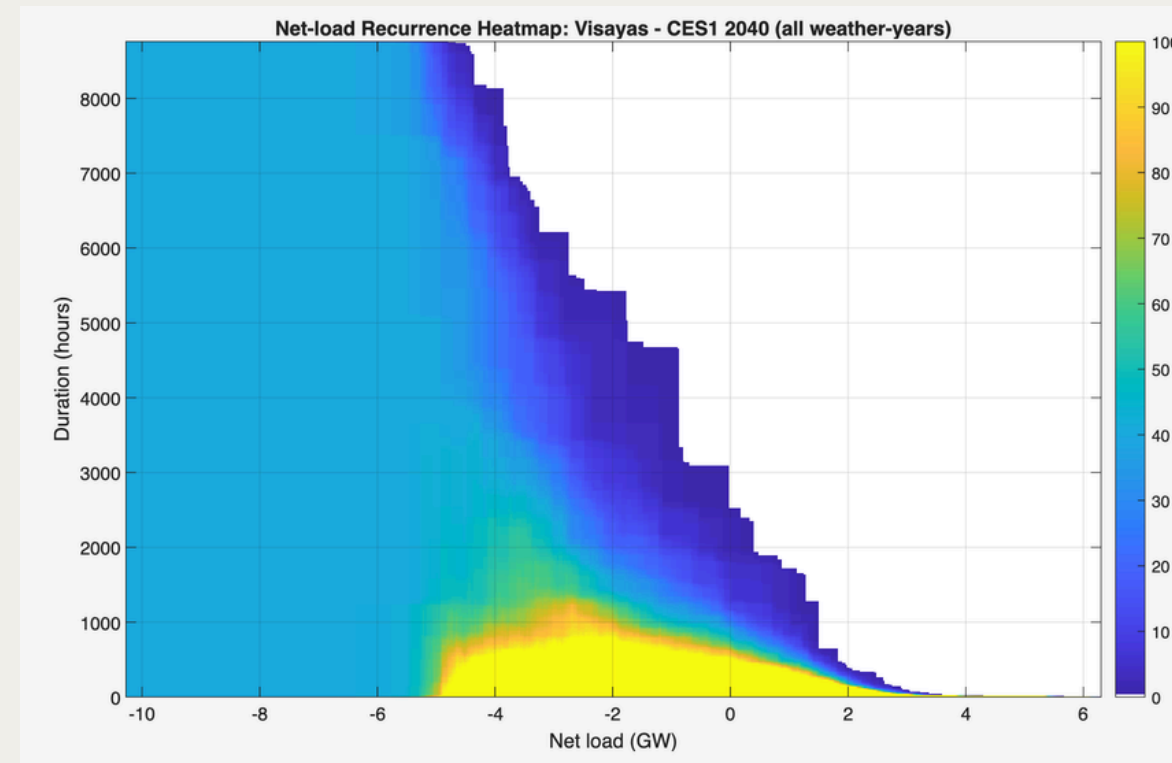
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Figure 8: 2035 CES 1 Recurrence Heatmap:
(a) Luzon, (b) Visayas, (c) Mindanao

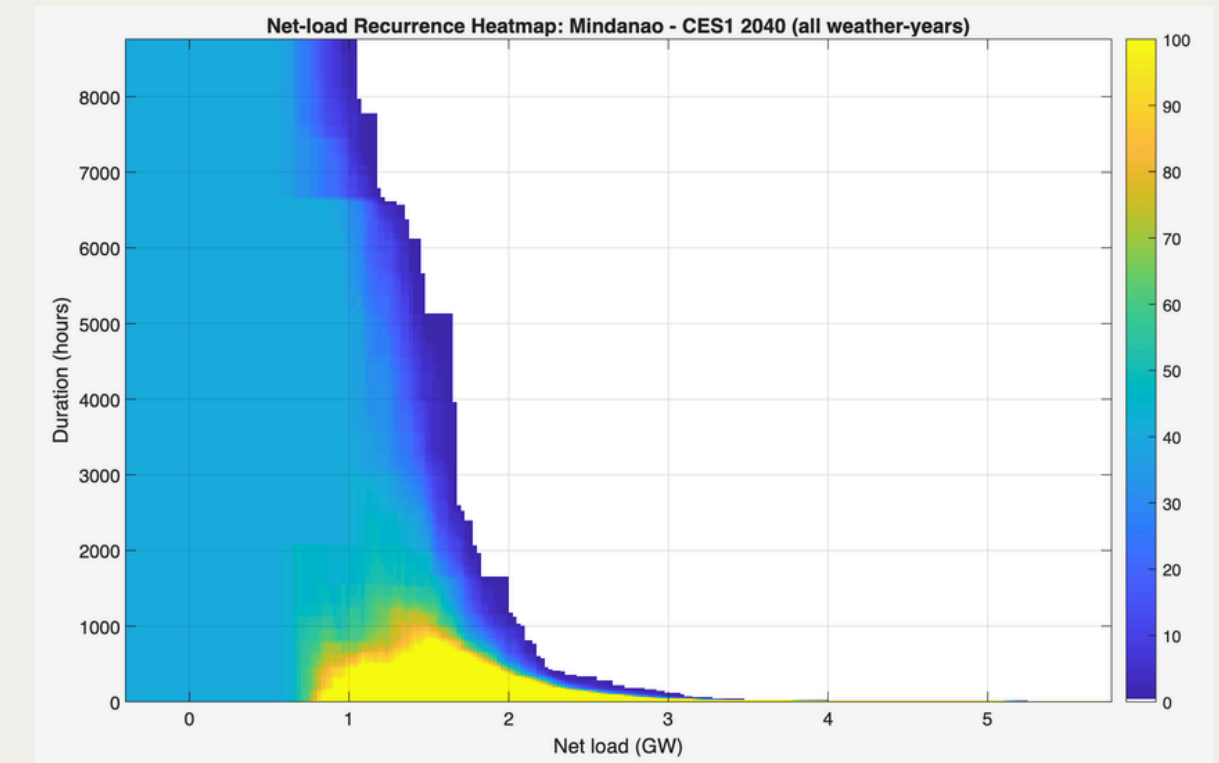
Heatmap: CES 1 (2040)



a.



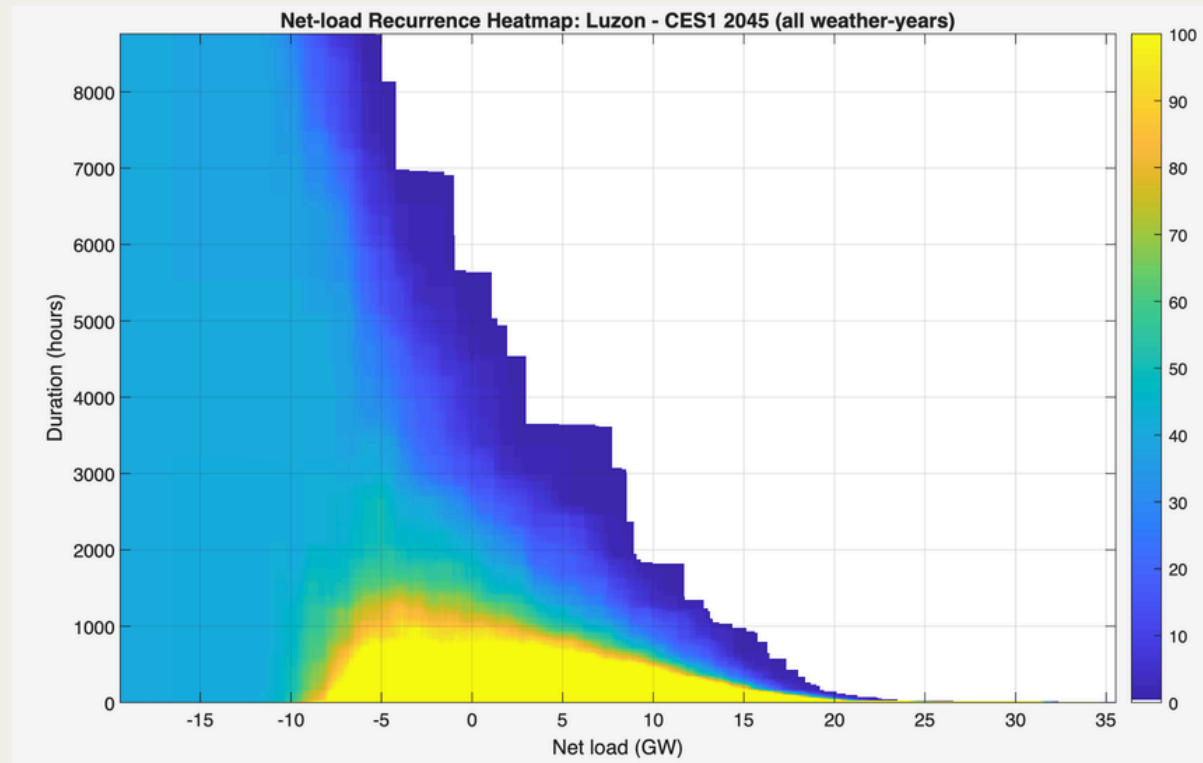
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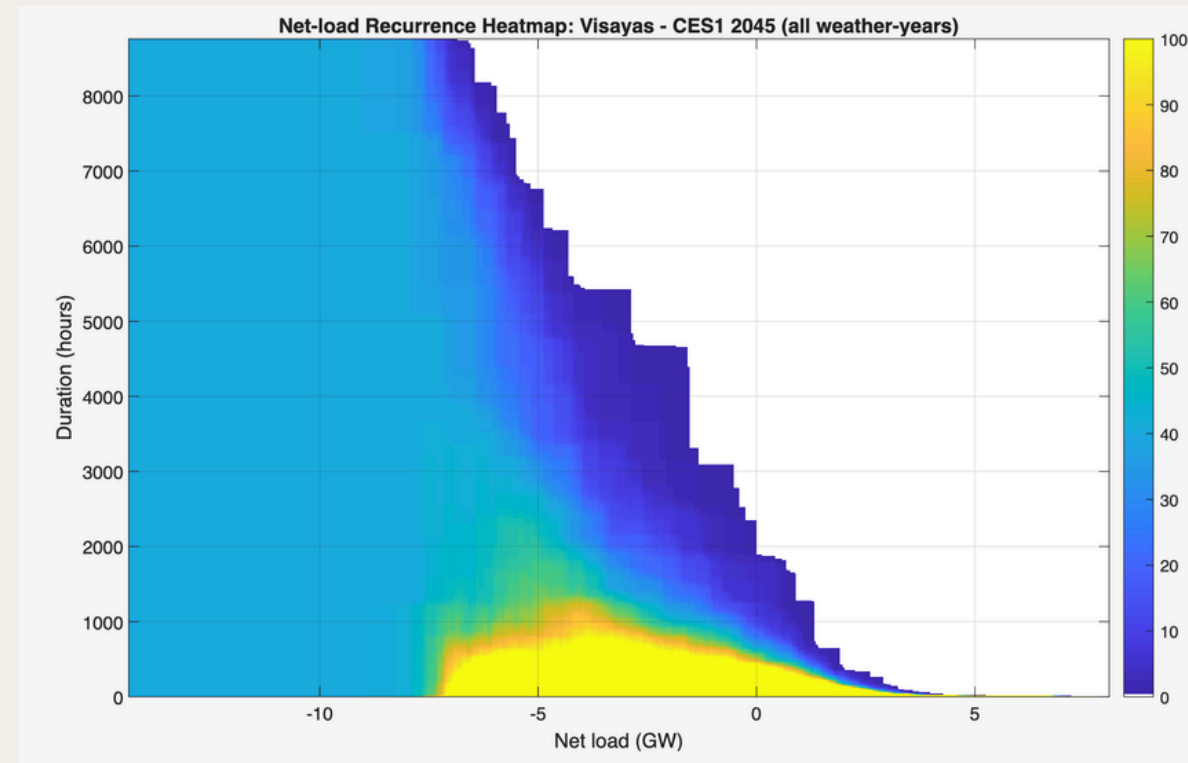
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Figure 9: 2040 CES 1 Recurrence Heatmap:
(a) Luzon, (b) Visayas, (c) Mindanao

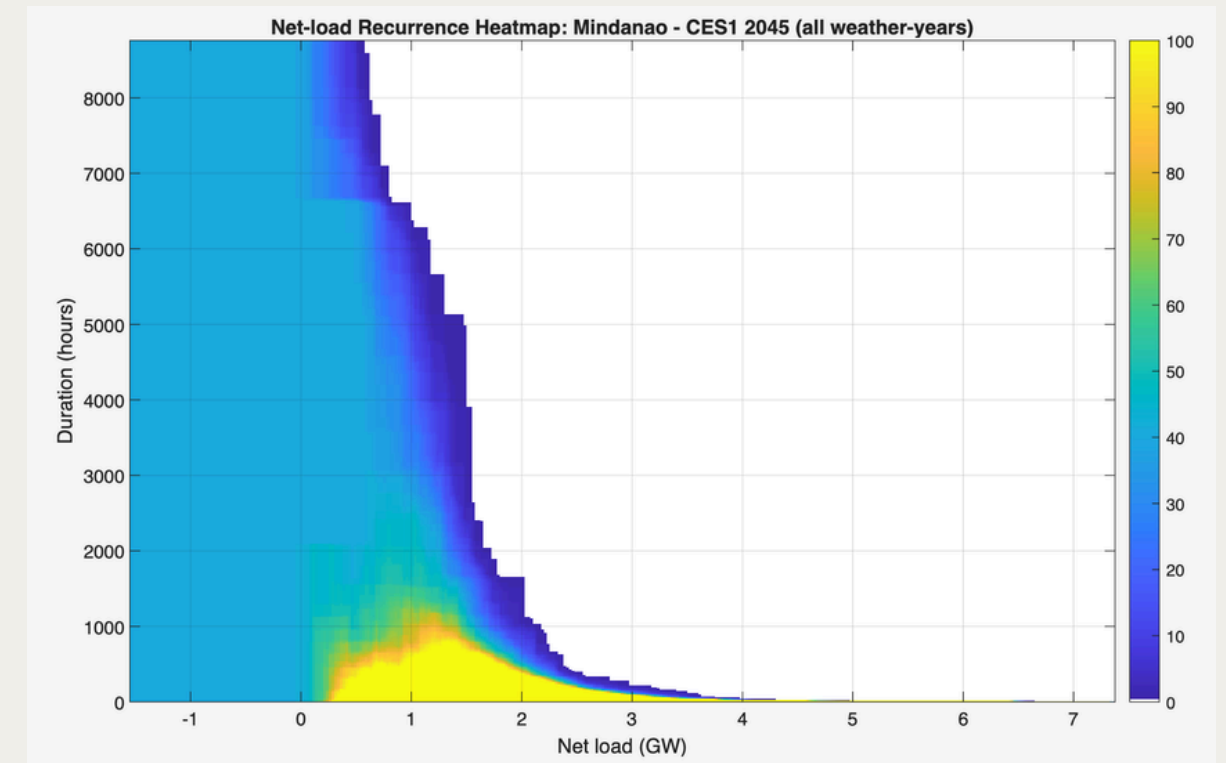
Heatmap: CES 1 (2045)



a.



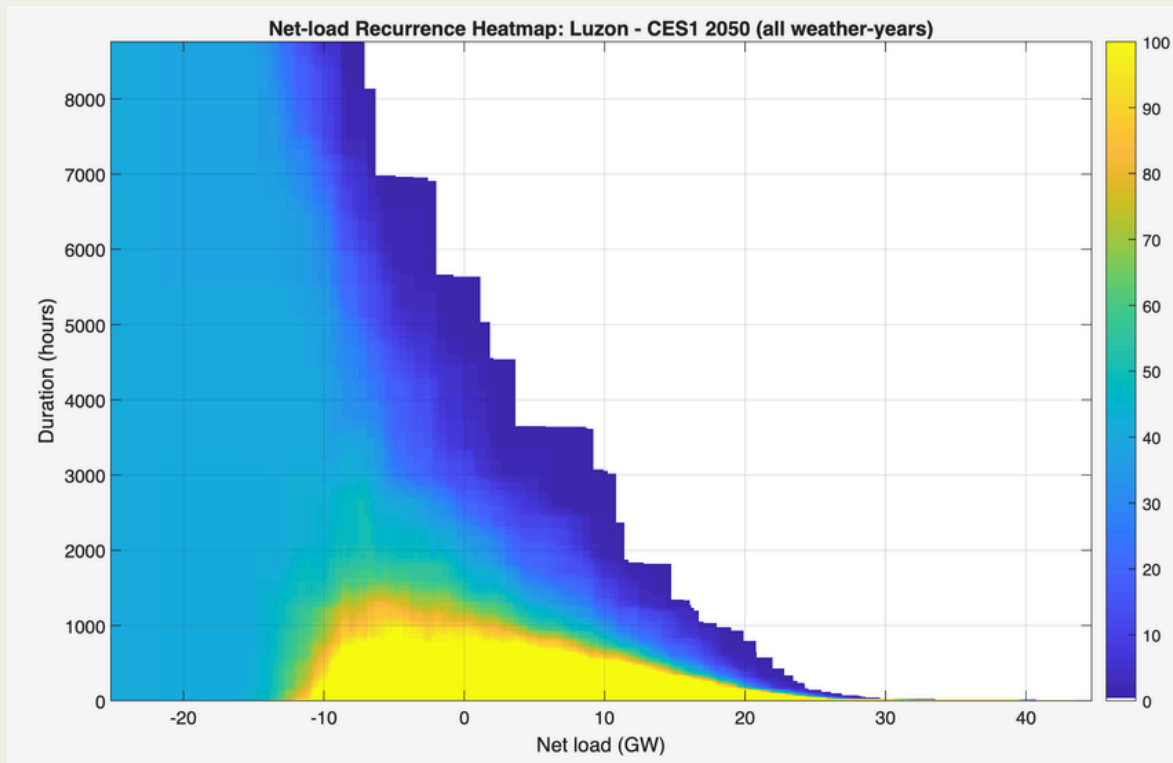
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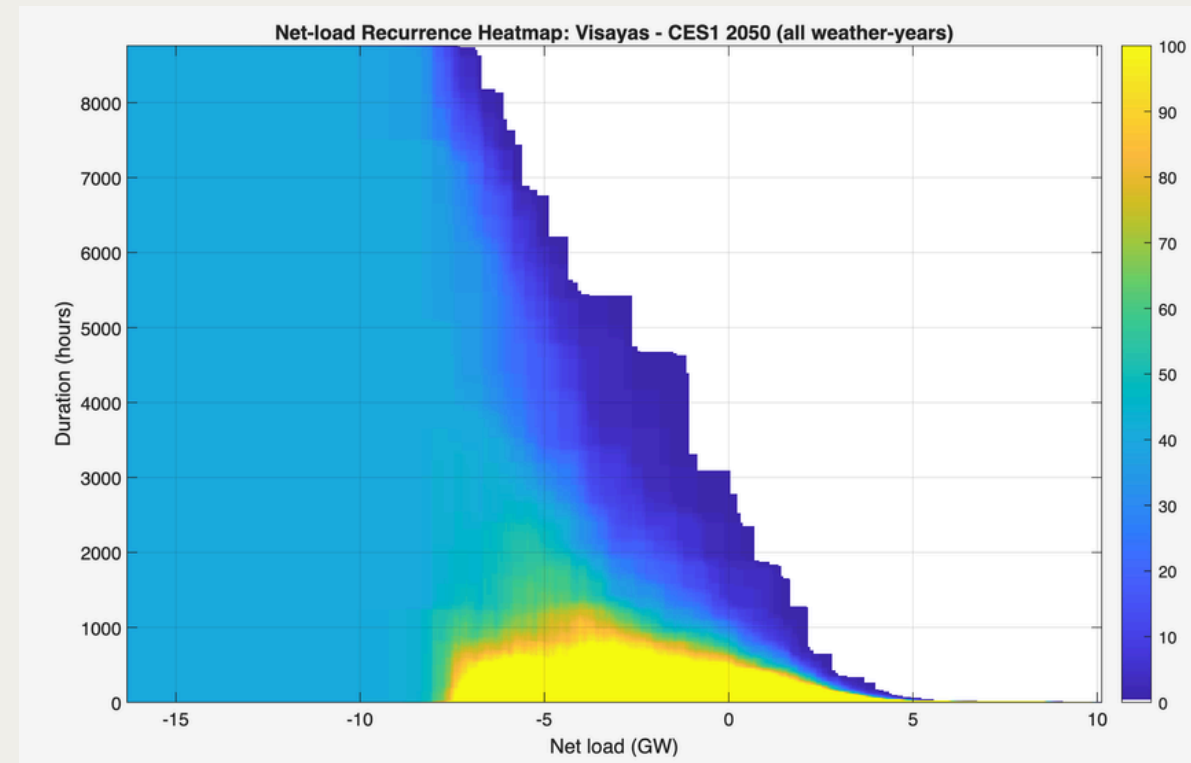
c.

Figure 10: 2045 CES 1 Recurrence Heatmap:
(a) Luzon, (b) Visayas, (c) Mindanao

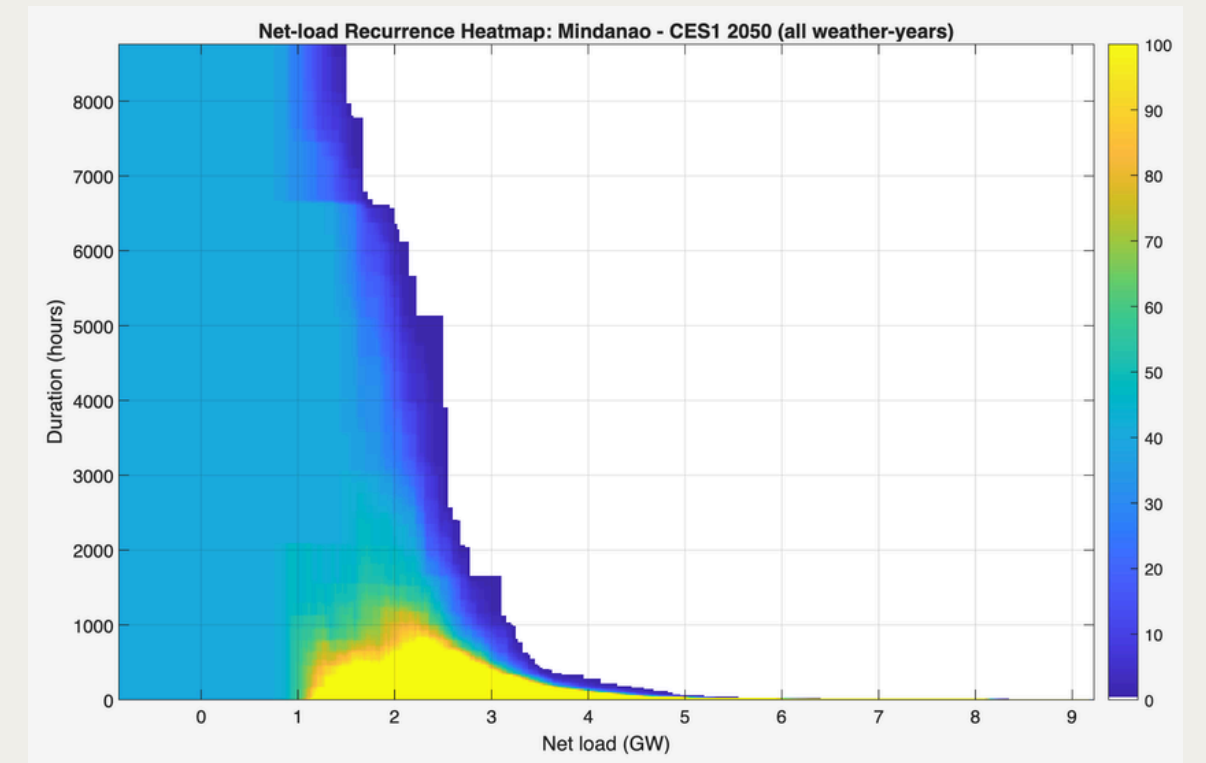
Heatmap: CES 1 (2050)



a.



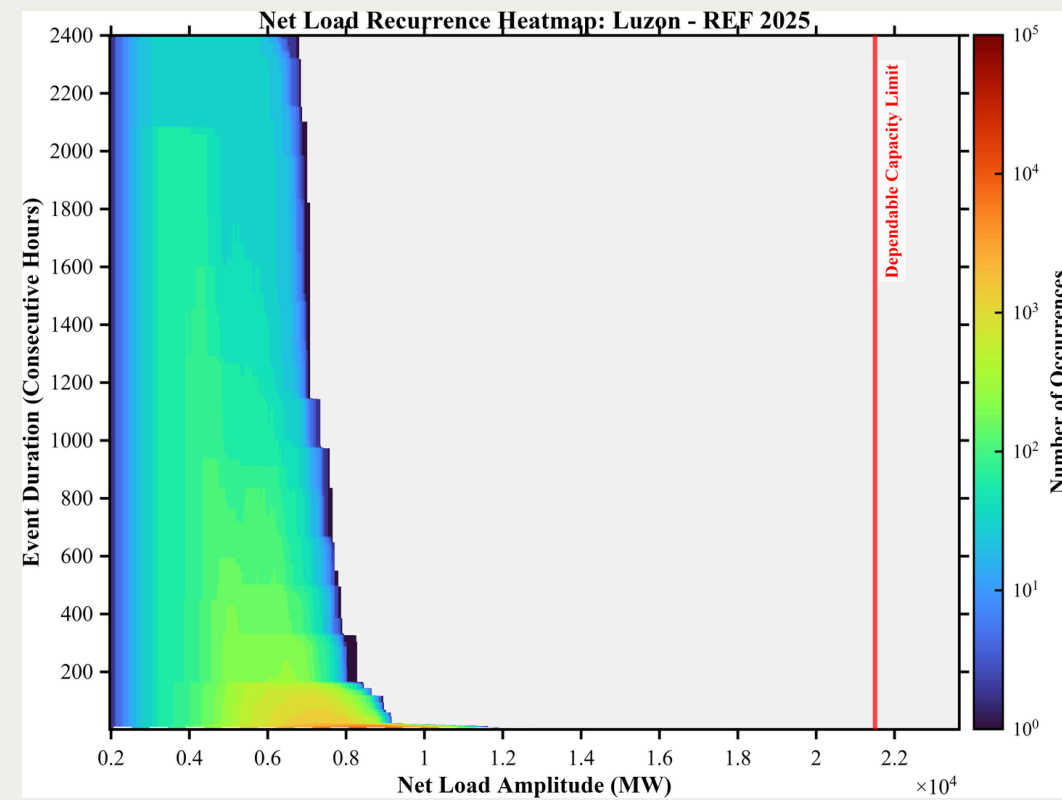
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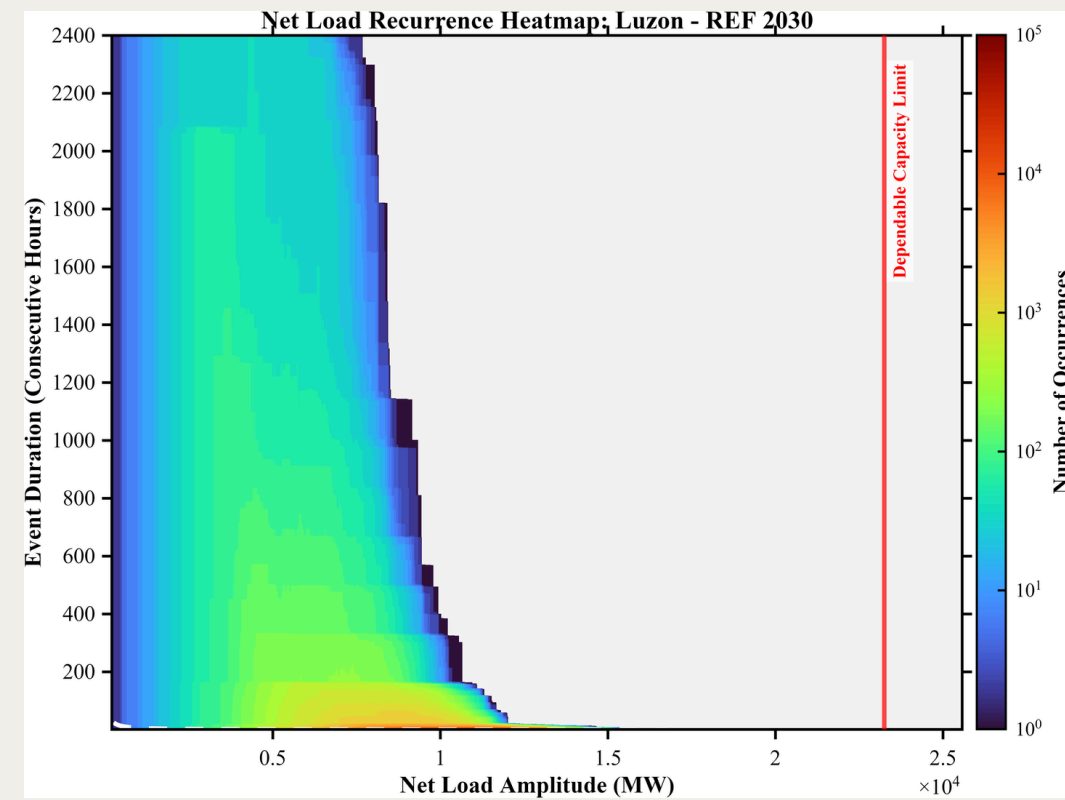
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Figure 11: 2050 CES 1 Recurrence Heatmap:
(a) Luzon, (b) Visayas, (c) Mindanao

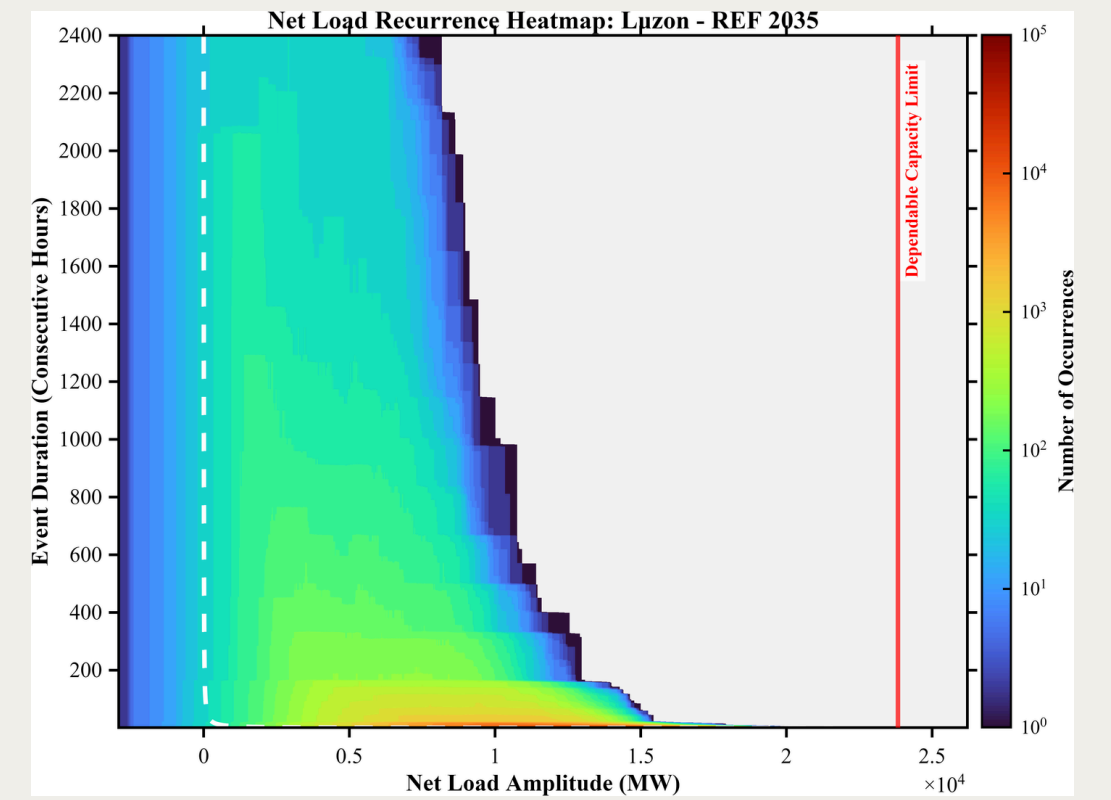
Luzon REF Capacity Stress Test



a.



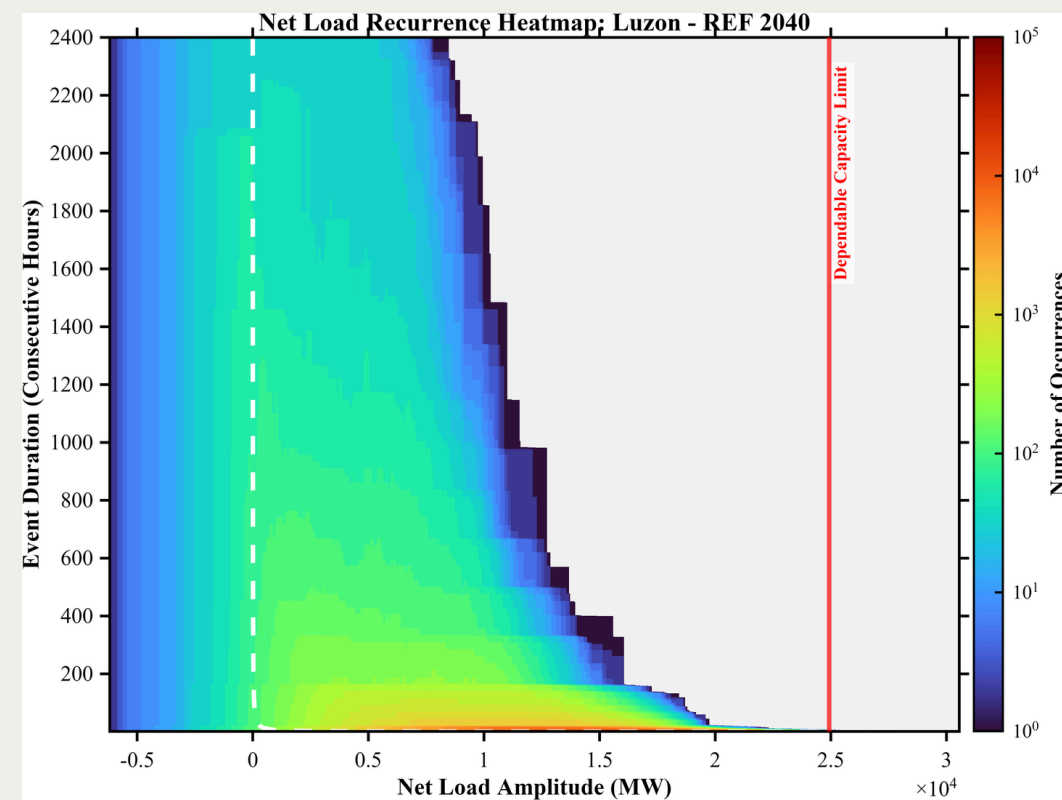
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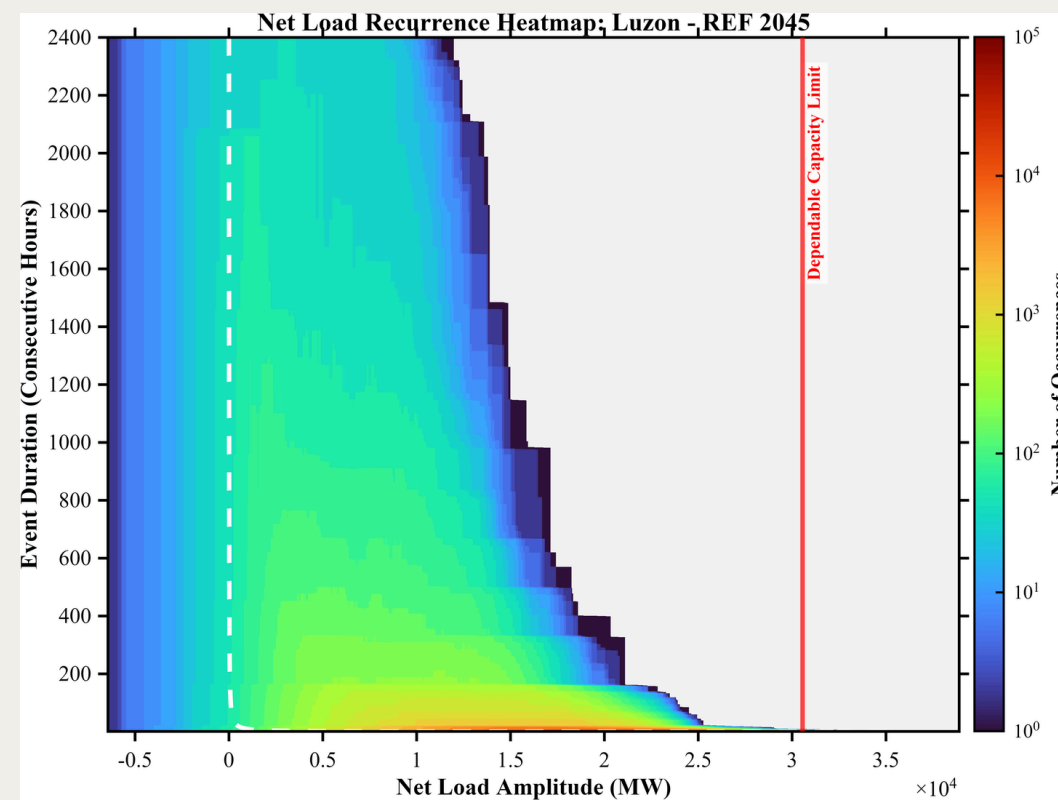
c.

Figure 11: Luzon REF Capacity Stress Test 1/2
(a) 2025, (b) 2030, (c) 2035

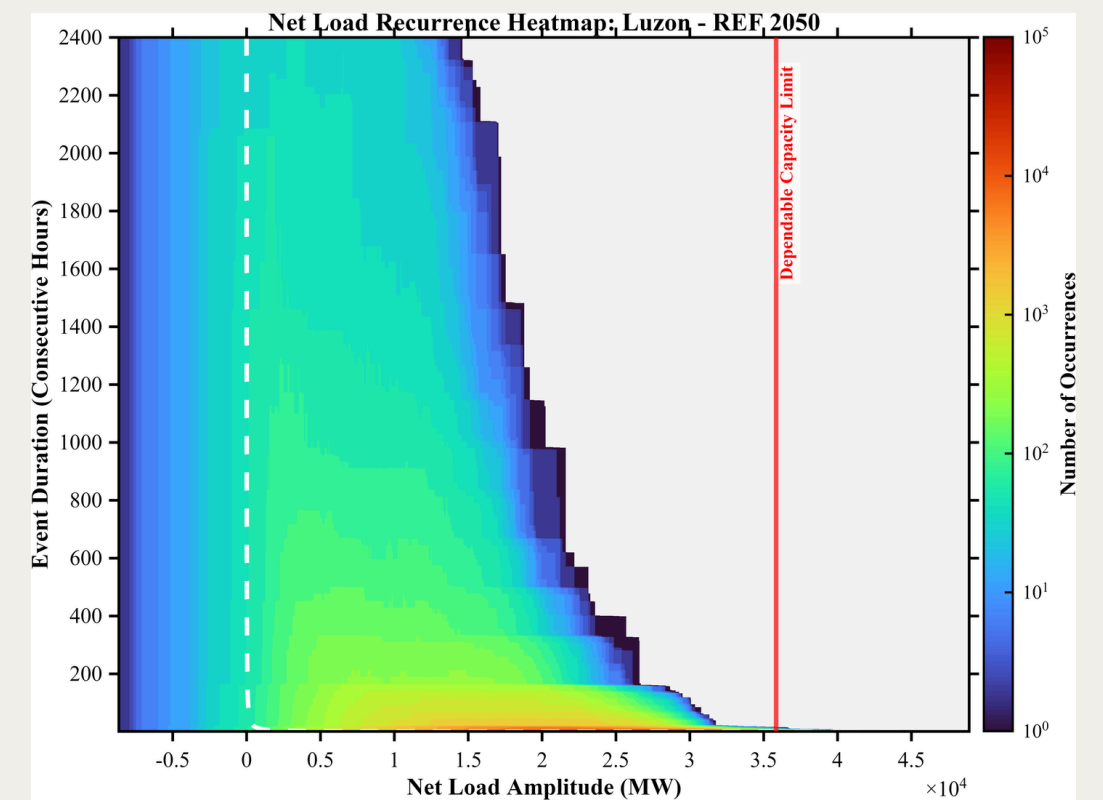
Luzon REF Capacity Stress Test



a.



b.



c.

Figure 11: Luzon REF Capacity Stress Test 2/2
(a) 2040, (b) 2045, (c) 2050

Remaining Tasks

- Proper Stress Tests
- Data Analysis
- Final Manuscript Writeup

Problems Encountered

- Delays in Hydro Data
- Different File and Data format for Nordic Study (trouble in replicating results)
- Negative Net Load
 - VRE > Peak Demand
 - Came from PDP Capacity Targets being greater than Total CREZ Capacity
 - Solution: Find proper model (where to put or how to model excess capacity)

Thank You!